

IBM Resiliency Summit

Future-Proofing Your Enterprise with GDPS

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Agenda



- Early solutions that exploited mirroring to provide DR
- Fateful catalyst for change
- New requirements for even higher levels of resilience
- What's the latest?
- GDPS Products / Features and Capabilities
- Summary



Early solutions that exploited mirroring to provide DR

Disaster Recovery prior to mirroring



Prior to mirroring, there was PTAM (Pickup Truck Access Method)



In the late 90's, disk mirroring was introduced



- Significantly reduced RPO – dictated by mirroring latency vs. backup frequency
- Significantly reduced RTO – hauling tapes to the DR site and restoring the entire environment from tape eliminated

GDPS Metro (initially known as GDPS/PPRC)

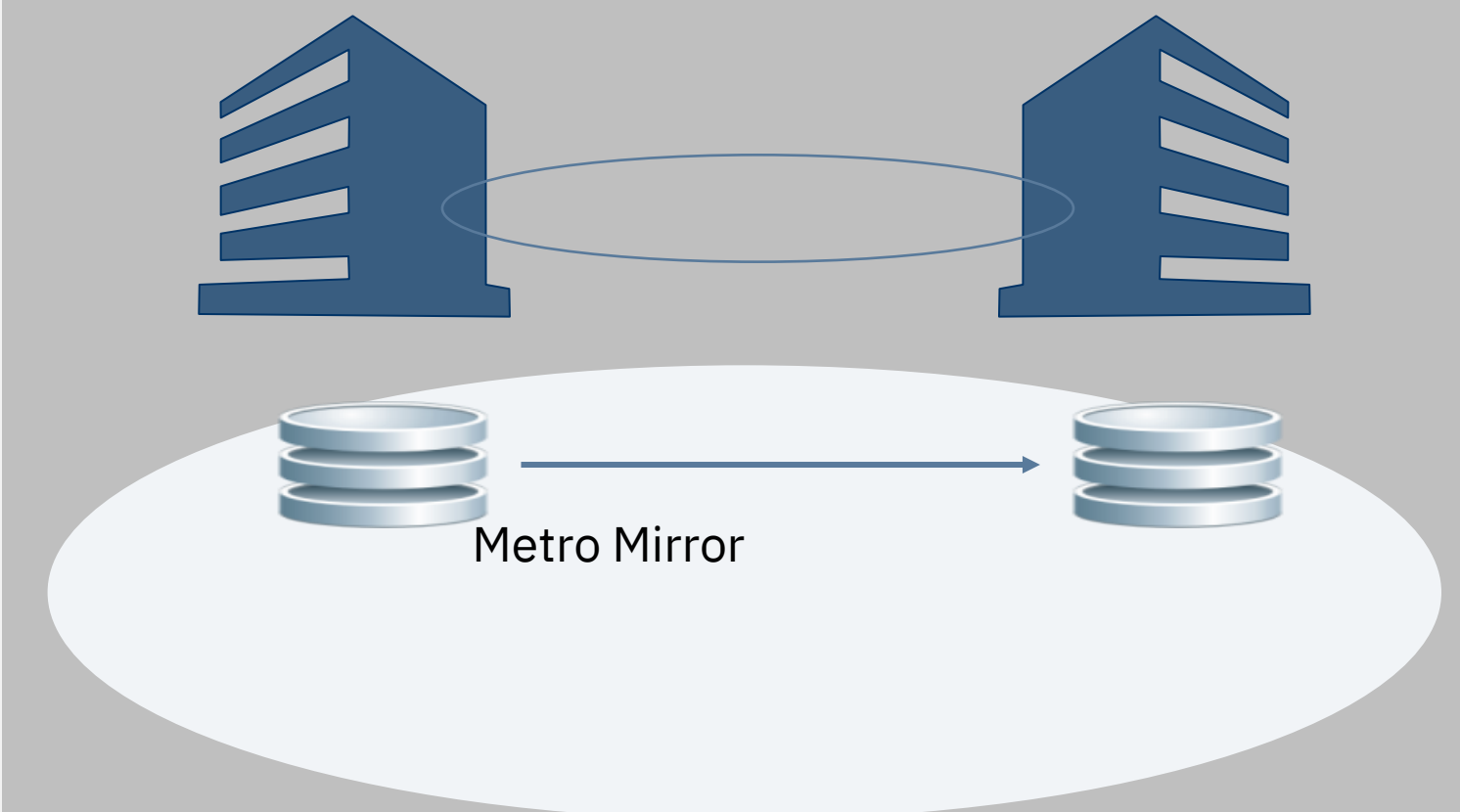


- Maintains two copies of data synchronously replicated
- Provides near-continuous availability and DR in a single data center or in a two data center environment within Metro Mirror distances
- Supports RTO of less than one hour (Single-Site Workload) and RPO of zero
- Provides autonomic recovery for primary disk failures and CEC failures
- Provides Sysplex and server management in addition to storage replication and recovery management
- Automates management of temporary capacity on demand (Flexible Capacity for Cyber Resiliency, CBU, OOCOD, CPE, BOOST)
- Provides FlashCopy capabilities for resync protection and offline backup
- Powerful scripting commands that highly automate recovery operations
- Monitors the environment and reports on events that could prevent rapid recovery

Continuous availability (CA) and disaster recovery (DR) within a metropolitan region

Two data centers

Multi-site workloads can withstand site and/or storage failures



RPO = 0 and RTO < 1 hour

GDPS Global (initially known as GDPS/XRC)



- Automates management of the Global Mirror technology
- Manages multiple consistency groups (scalable)
- Automates recovery of the IBM Z applications at the recovery site
- Supports RTO of less than one hour and a RPO as low as two seconds *
- Uses FlashCopy management for data protection and to facilitate in demand DR
- Automates management of temporary capacity on demand (Flexible Capacity for Cyber Resiliency, CBU, OOCOD, CPE, BOOST)
- Powerful scripting commands that highly automate recovery operations
- Monitors the environment and reports on events that could prevent rapid recovery

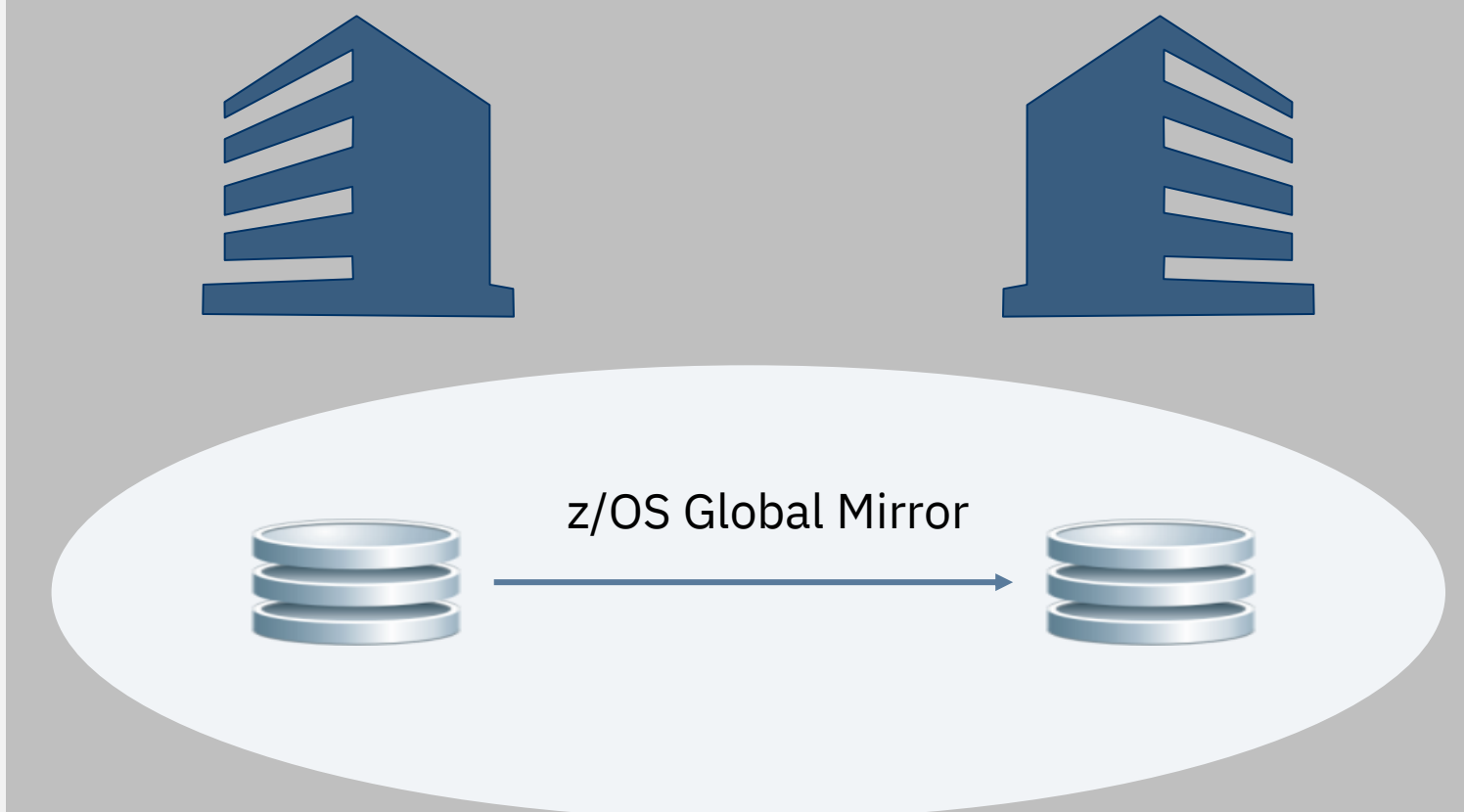
*RPO (Data loss) is workload and bandwidth-dependent

Disaster Recovery at Extended Distance

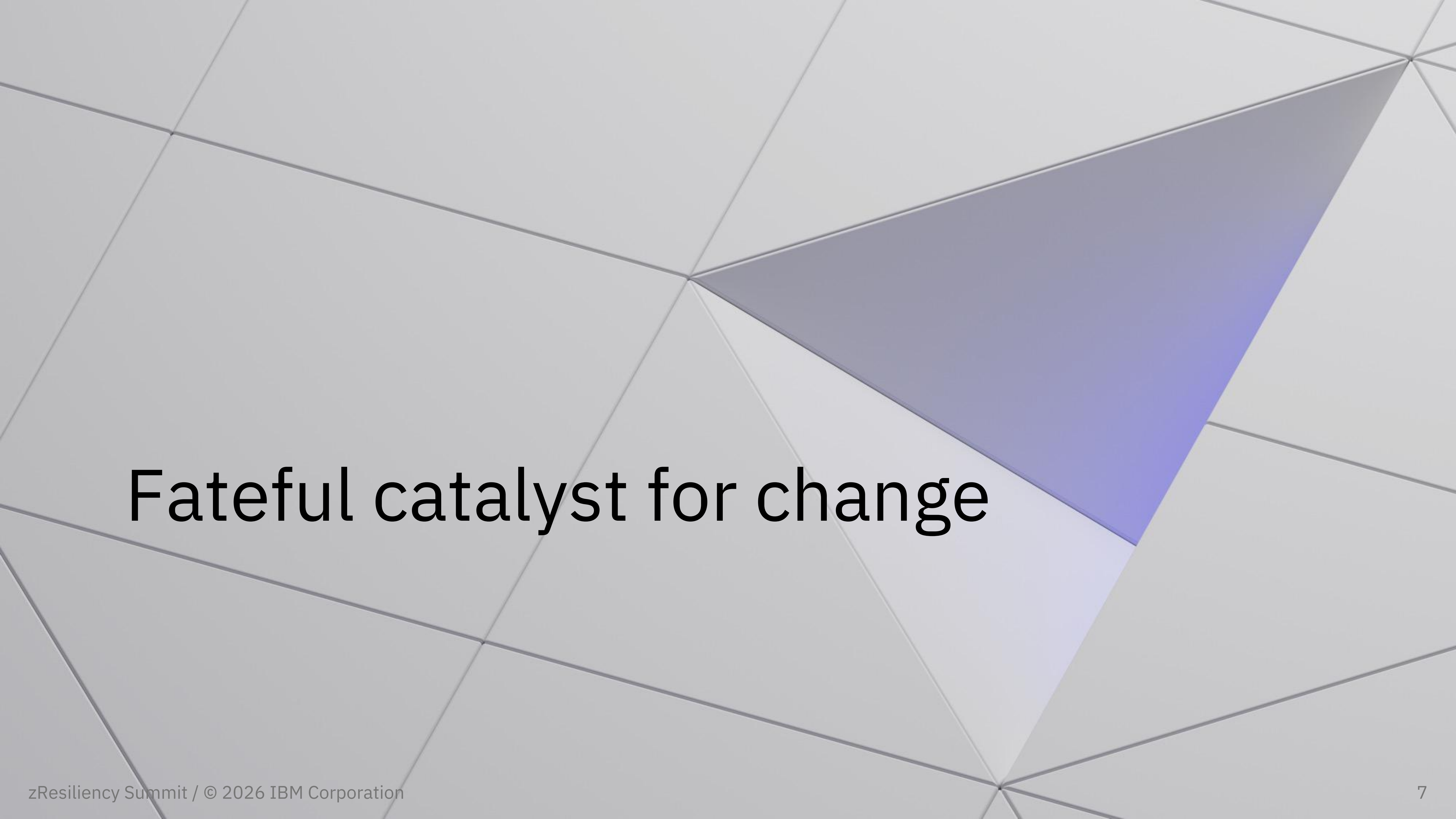
Two Data Centers

Rapid Systems Disaster Recovery with
“seconds” of Data Loss

Disaster recovery for out of region interruptions



RPO = 2-5 secs, RTO = an hour or less

The background features a light gray field with a network of thin, dark gray lines intersecting at various points. A large, prominent triangle is filled with a gradient of blue, transitioning from a lighter shade at the bottom to a darker shade at the top. The text "Fateful catalyst for change" is centered in a large, black, sans-serif font.

Fateful catalyst for change

Fateful catalyst for change



- Sept 11, 2001 terrorism and Interagency Paper on Sound Practices to Strengthen the Resilience of the U.S. Financial System [Docket No. R-1128] (April 7, 2003):
 - *Maintain sufficient geographically dispersed resources to meet recovery and resumption objectives*
 - *Routinely use or test recovery and resumption arrangements*
- Large majority of GDPS clients used GDPS Metro across two metro sites, many within 10 KMs due to ESCON and sysplex timer limitations
- *The GDPS configuration that clients deploy largely depends upon the data center location(s) and associated risk exposures*

GDPS Metro single site workload & multi-site workload configurations (aka MM 2-site or MM 3-site)*

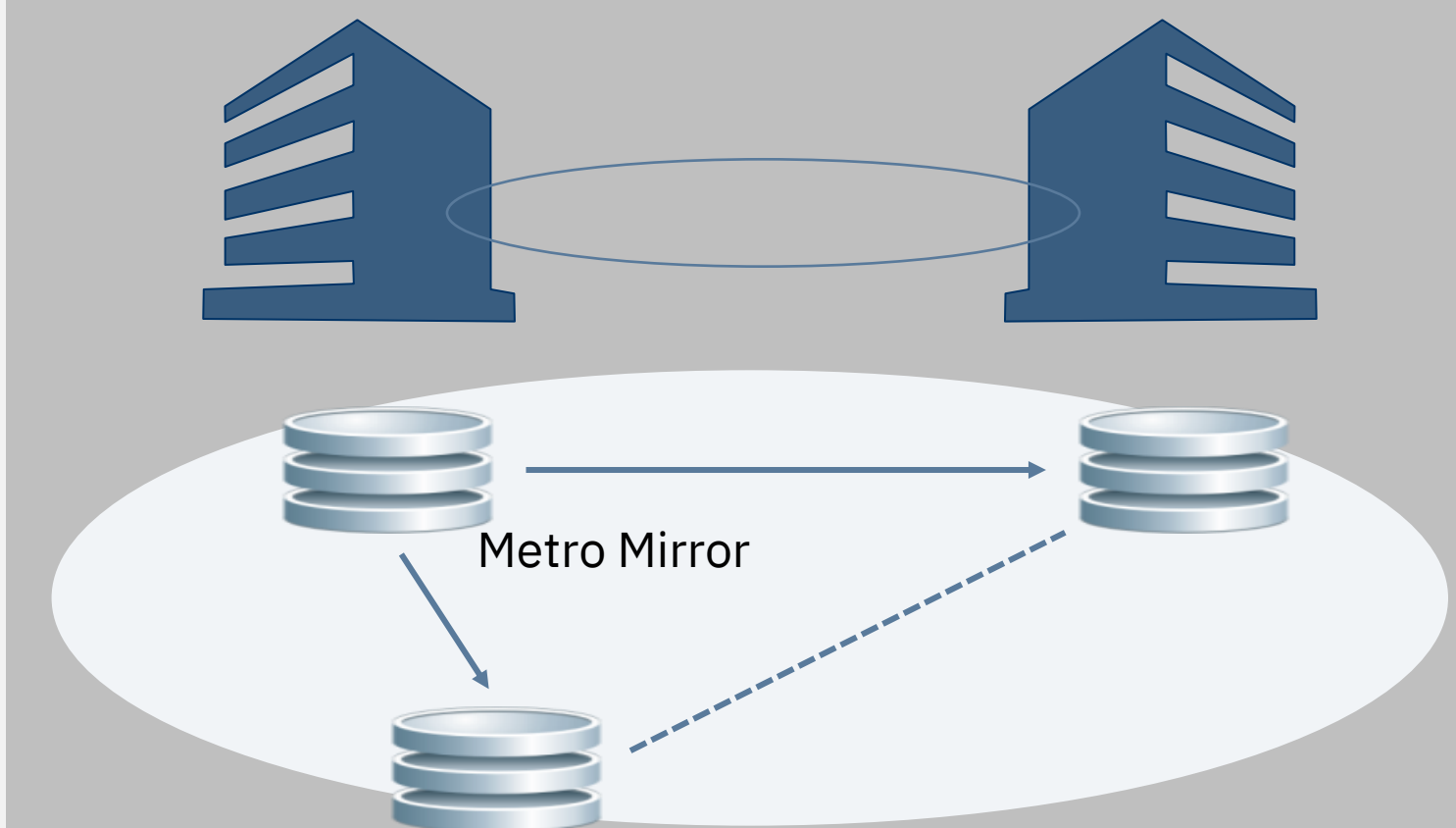


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- Provides autonomic recovery for primary disk failures and CEC failures
- Provides Sysplex and server management in addition to storage replication and recovery management
- Automates management of temporary capacity on demand (Flexible Capacity for Cyber Resiliency, CBU, OOCOD, CPE, BOOST)
- Provides FlashCopy capabilities for resync protection and offline backup
- *Single-site workloads and multi-site workloads supported*
- Powerful scripting commands that highly automate recovery operations
- Monitors the environment and reports on events that could prevent rapid recovery

Continuous availability (CA) and disaster recovery (DR) within a metropolitan region

**Two/three data centers
(2 server sites, 3 disk locations)**

Multi-site workloads can withstand site and/or storage failures



RPO = 0 and RTO ≈ 2 mins / < 1 hour

*distance between sites increased to 100 fiber KM due to new protocols (FICON & PPRC over fiber), STP, & DWDM

GDPS Metro Global – GM 3-site (aka MGM 3-site)

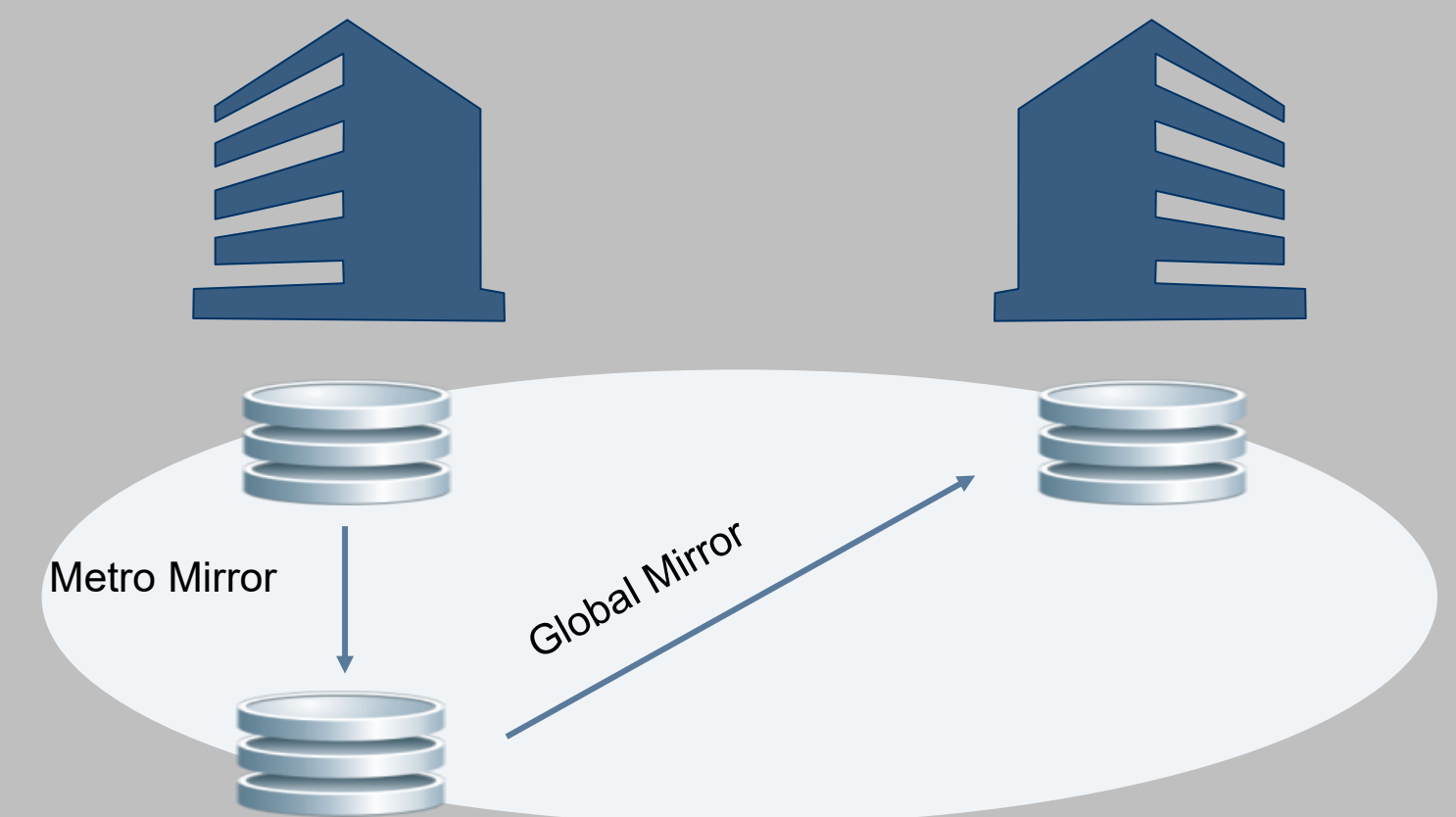


- Provides continuous availability and long-distance DR in a two or three data center environment
- Supports RTO of less than one hour and RPO of zero for local failure events
- Supports RTO of less than one hour and RPO as low as two seconds for regional failure events
- Monitors and manages all sites from a single point of control
- Powerful scripting commands that highly automate recovery operations
- Automated DR testing with minimal impact to RPO

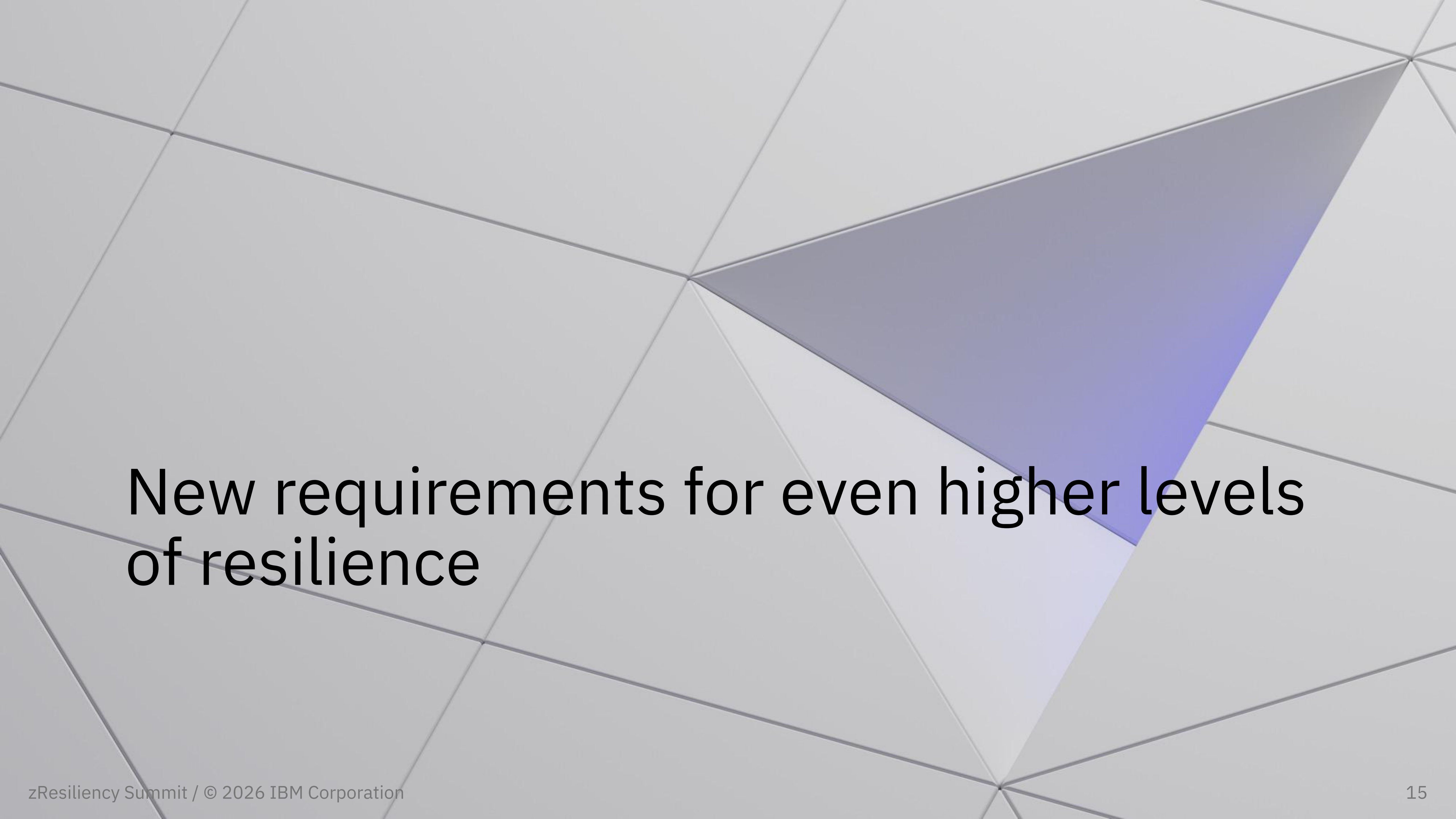
**Continuous availability (CA) locally
and DR at extended distances**

Two or three data centers

**High availability for site disasters
Disaster recovery (DR) for regional disasters**



RPO = 2-5 secs, RTO = an hour or less



New requirements for even higher levels of resilience

Requirements become more stringent



- Clients hesitant to declare DR...
 - DR testing doesn't provide sufficient confidence that production can actually be run in the DR region under peak workloads
- Financial services sector clients being encouraged by regulators perform regular region toggles that verify DR capability, that prove production can run in the DR region under peak workloads, and that demonstrates ability to return home after a DR event...Region Toggle (aka “switch and stay”)
- *Clients, regardless of the GDPS configuration, are performing site / region switching*
 - *Site switching – switching z/OS images within a sysplex spread across two sites*
 - *Region switching – switching sysplexes between two regions*

GDPS Metro single site workload & multi-site workload configurations (aka MM 2-site or MM 3-site)

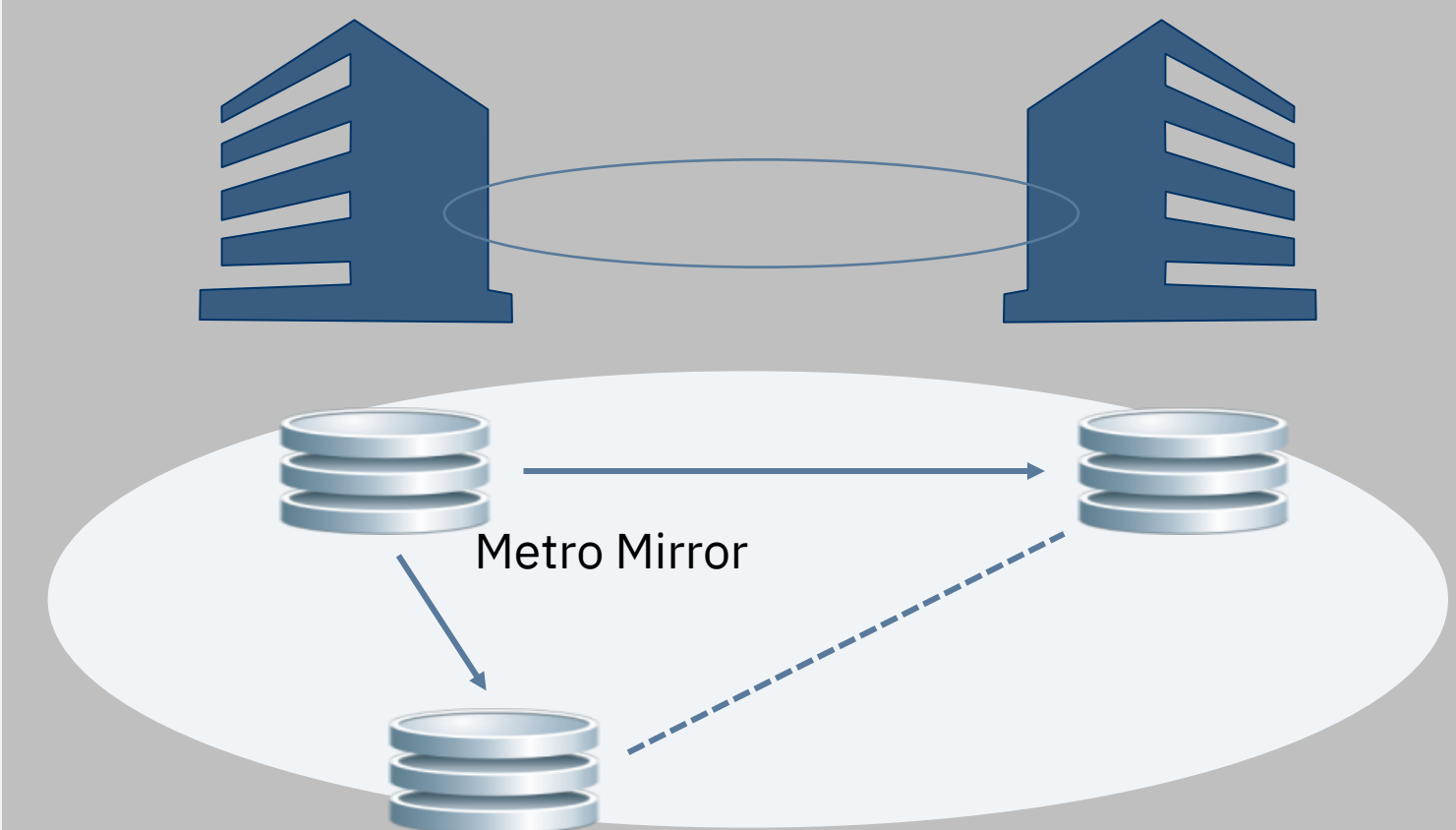


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- Automates management of temporary capacity on demand (Flexible Capacity for Cyber Resiliency, CBU, OOCOD, CPE, BOOST)
- Provides FlashCopy capabilities for resync protection and offline backup
- Single-site workloads and multi-site workloads
- *Site switching capability (prereqs sysplex enabled applications)*
- *Protects Linux on IBM Z environments (z/VM, KVM, and native in LPAR) as well as monoplexes and IBM Db2 Analytics Accelerator*
- Powerful scripting commands that highly automate recovery operations
- Monitors the environment and reports on events that could prevent rapid recovery

Continuous availability (CA) and disaster recovery (DR) within a metropolitan region

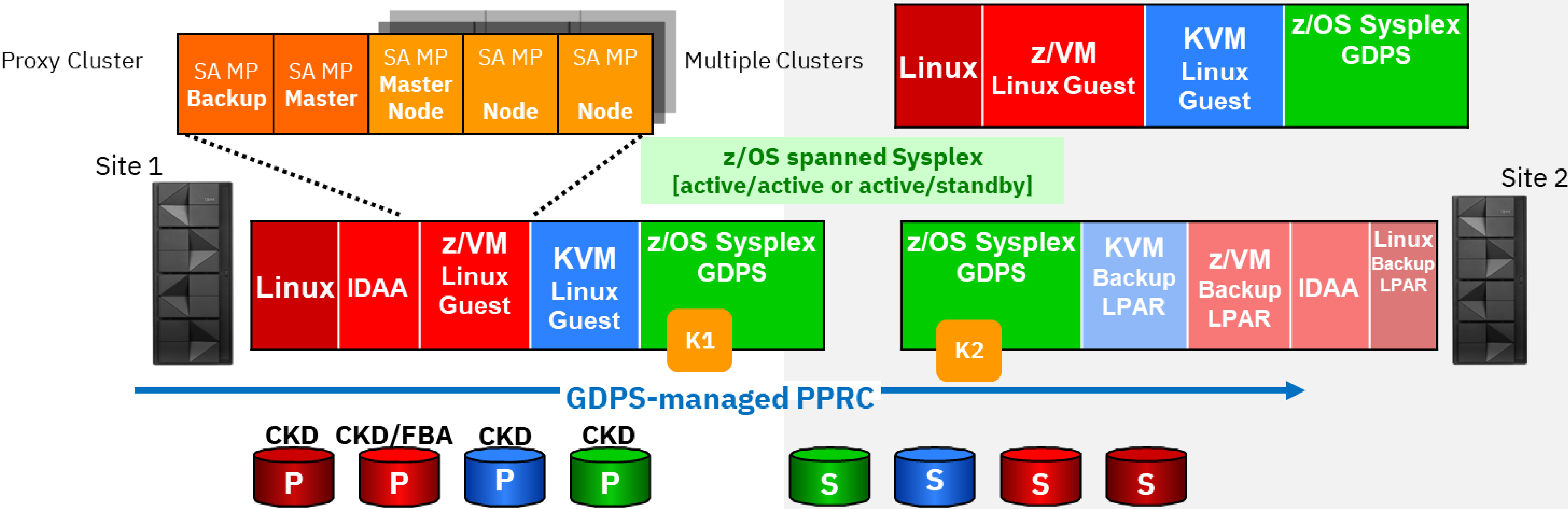
**Two/three data centers
(2 server sites, 3 disk locations)**

Multi-site workloads can withstand site and/or storage failures



RPO = 0 and RTO $\approx$ 2 mins / < 1 hour

GDPS End-to-End (xDR) for IDAA, zVM, zKVM, & LoZ



- Provides for high availability and continuous operations within one site across two or at global distances with near-continuous access to data
 - Automates management of the IBM Metro Mirror Technology
 - Automates HyperSwap for zVM and its guests & HyperSwap Restart for KVM with its guests
 - Supports zVM SSI Live Guest Relocation
- Management of all LPARs → start, stop, monitor...
 - Graceful shutdown and startup (re-IPL in place) of z/VM guests
 - Coordinated takeover in unplanned cases e.g. recovery from a node failure
 - Coordinated takeover in planned cases for e.g. maintenance
 - Can enable high availability and disaster recovery for Linux on IBM Z environments

Switch and stay in a GDPS Metro Global – GM 3-site not desirable

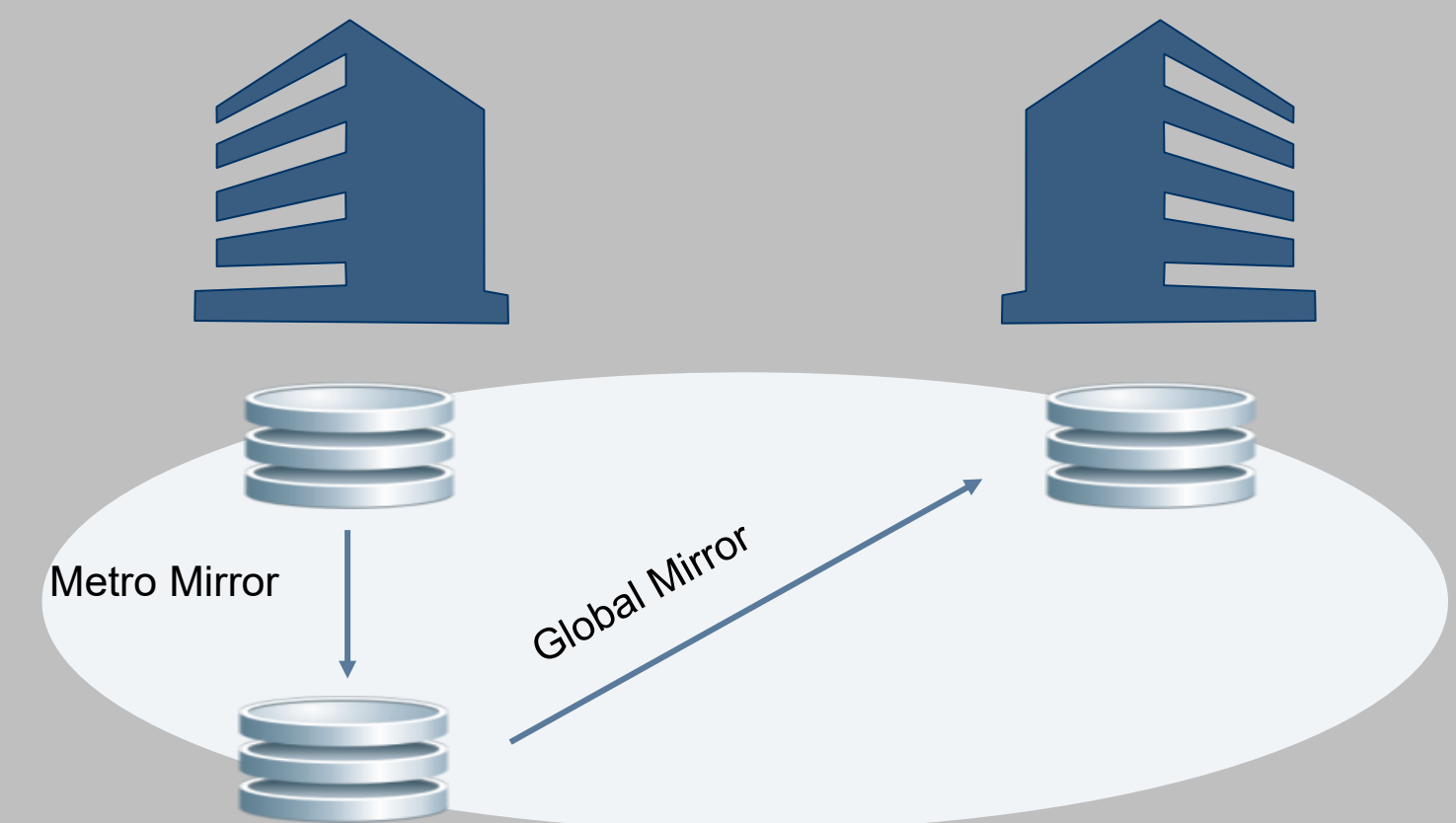


- DR data center does not provide the same level of resilience as the production data center
 - Only one copy of data, no HA for storage
- Different operational procedures
 - GDPS Metro running and managing systems and sysplex resources in production data center, no GDPS Metro running in the DR data center
- No DR protection while running in the DR data center

Continuous availability (CA) locally and DR at extended distances

Two or three data centers

**High availability for site disasters
Disaster recovery (DR) for regional disasters**



RPO = 2-5 secs, RTO = an hour or less

GDPS Metro Global – GM 4-site (aka MGM 4-site)



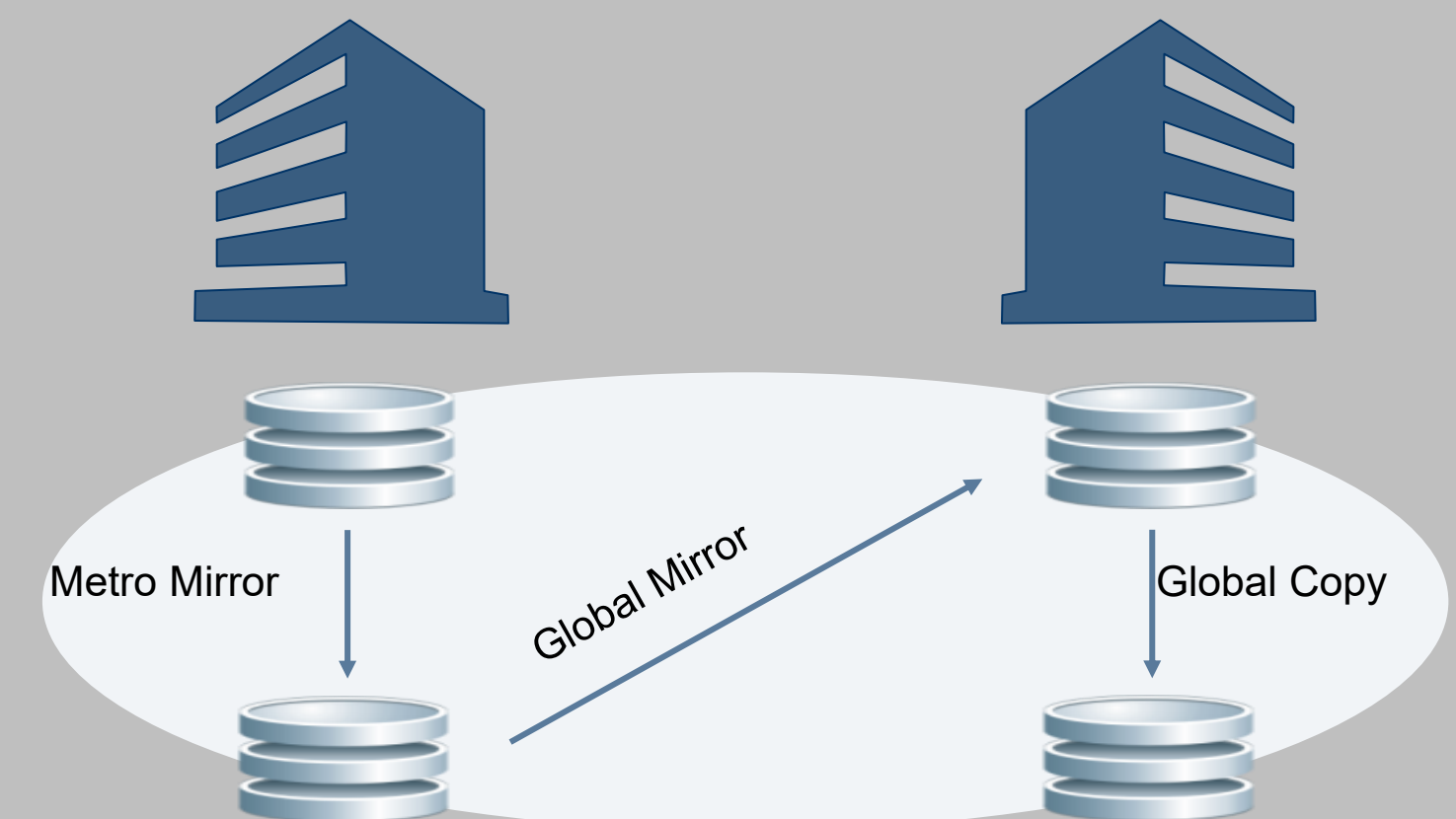
- Provides continuous availability and long-distance DR in a two, three, or four data center environment
- Supports RTO of less than one hour and a RPO as low as two seconds*
- Monitors and manages all sites from a single point of control
- Powerful scripting commands that highly automate recovery operations
- Symmetric configuration provides same levels of resilience and same operational procedures in both regions – facilitating region toggle (“switch and stay”)

*RPO (Data loss) is workload and bandwidth-dependent

Continuous availability (CA) locally and DR at extended distances

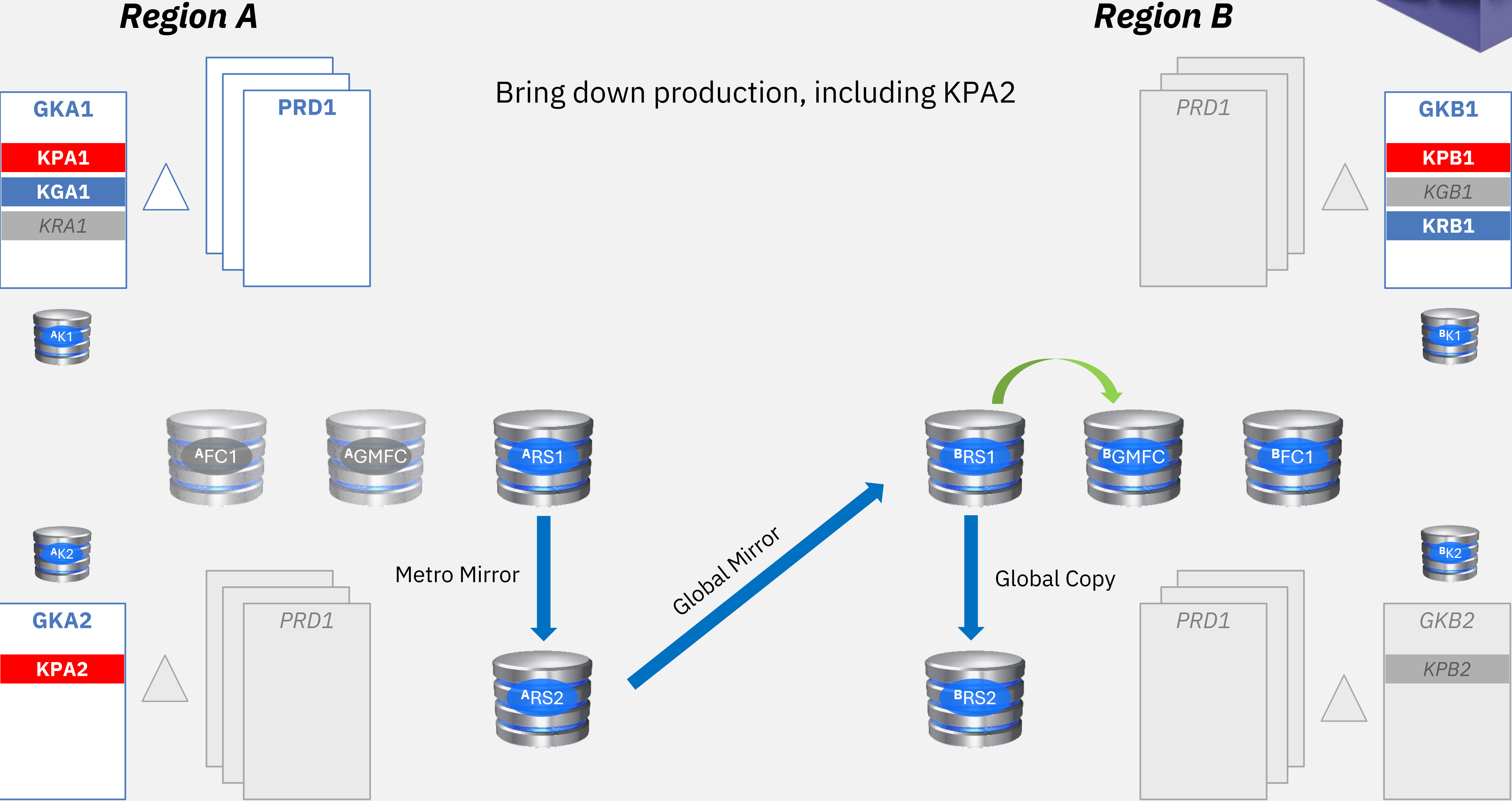
Two, three, or four data centers

High availability for site disasters
Disaster recovery (DR) for regional disasters

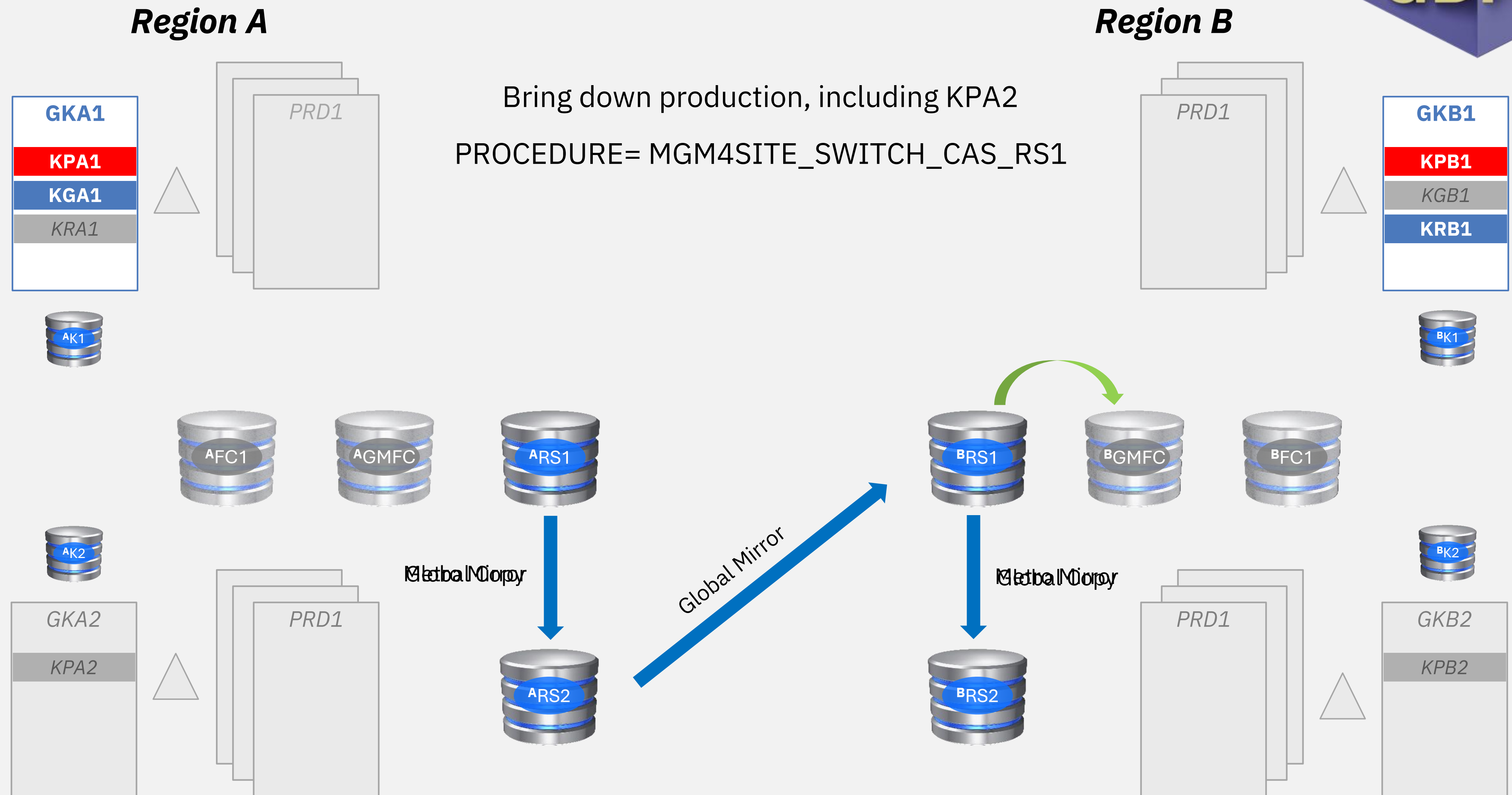


RPO = 2-5 secs, RTO = an hour or less

Planned region switch - MGM 4-site



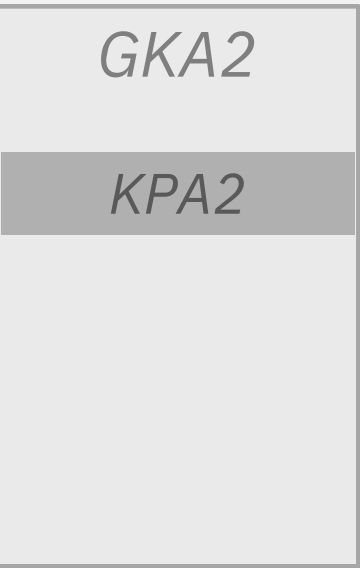
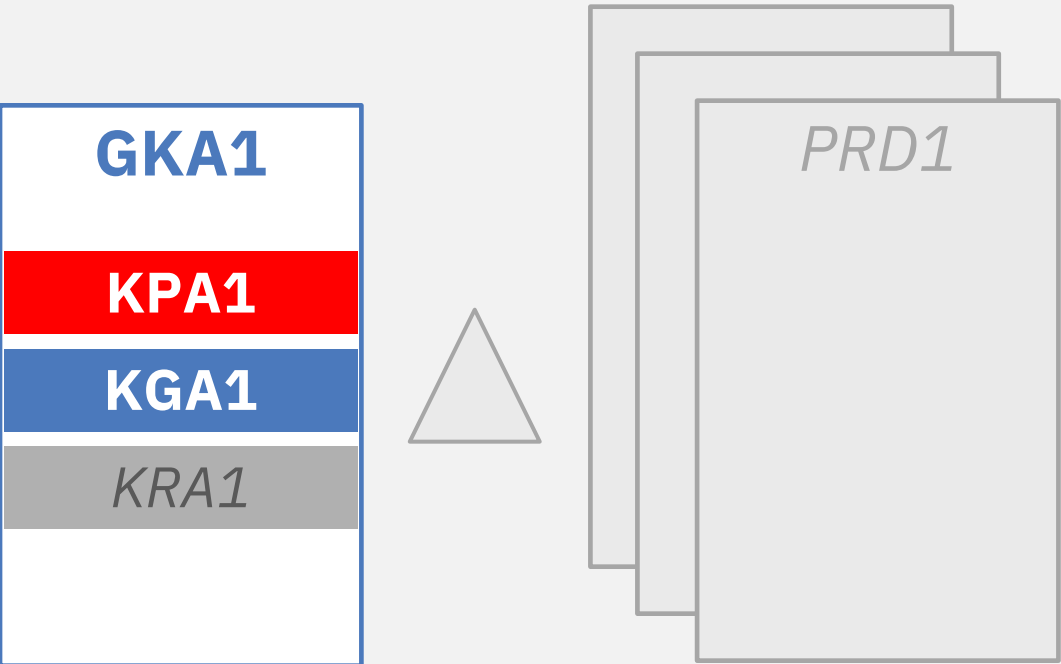
Planned region switch - MGM 4-site



Planned region switch - MGM 4-site

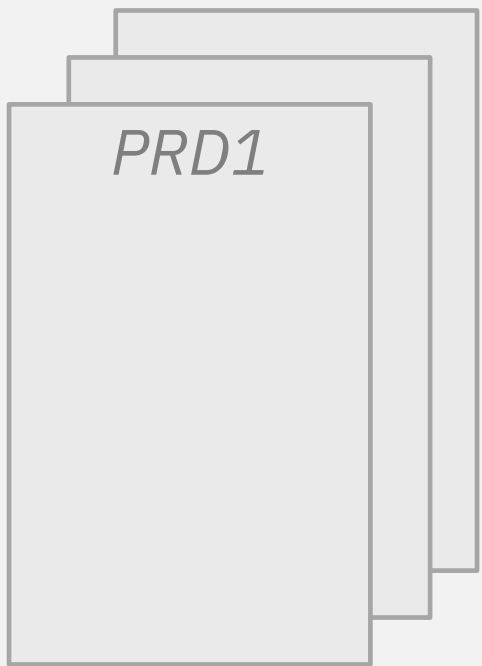
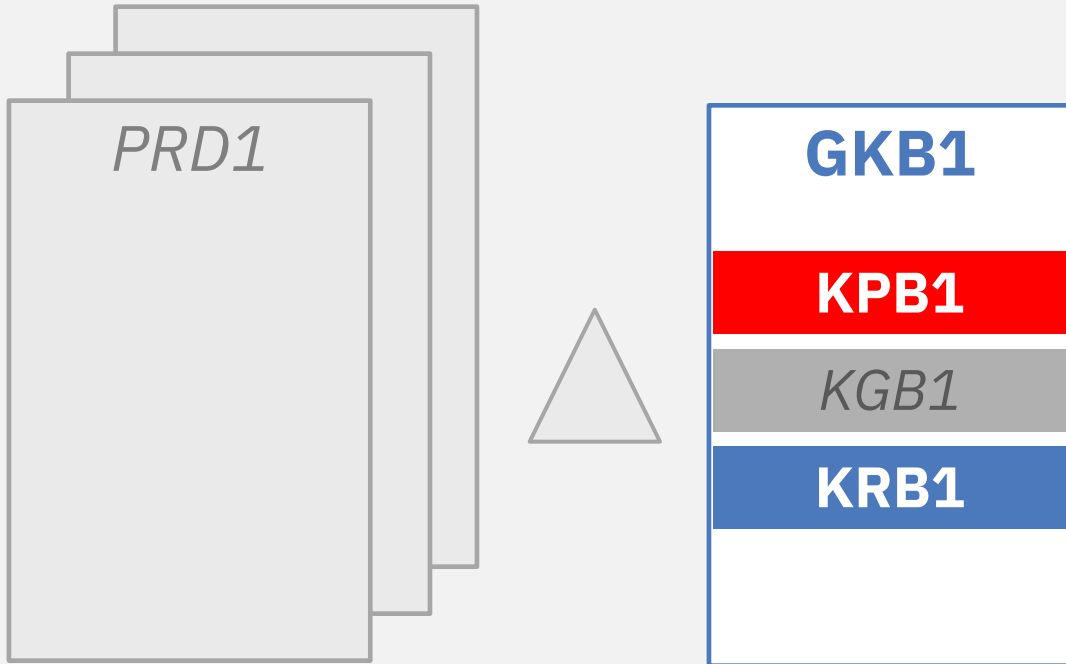


Region A



Bring down production, including KPA2
PROCEDURE= MGM4SITE_SWITCH_CAS_RS1
GDASD = 'INITIALIZE'

Region B



Global Copy



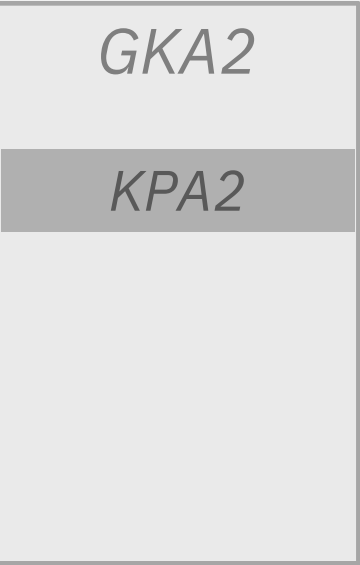
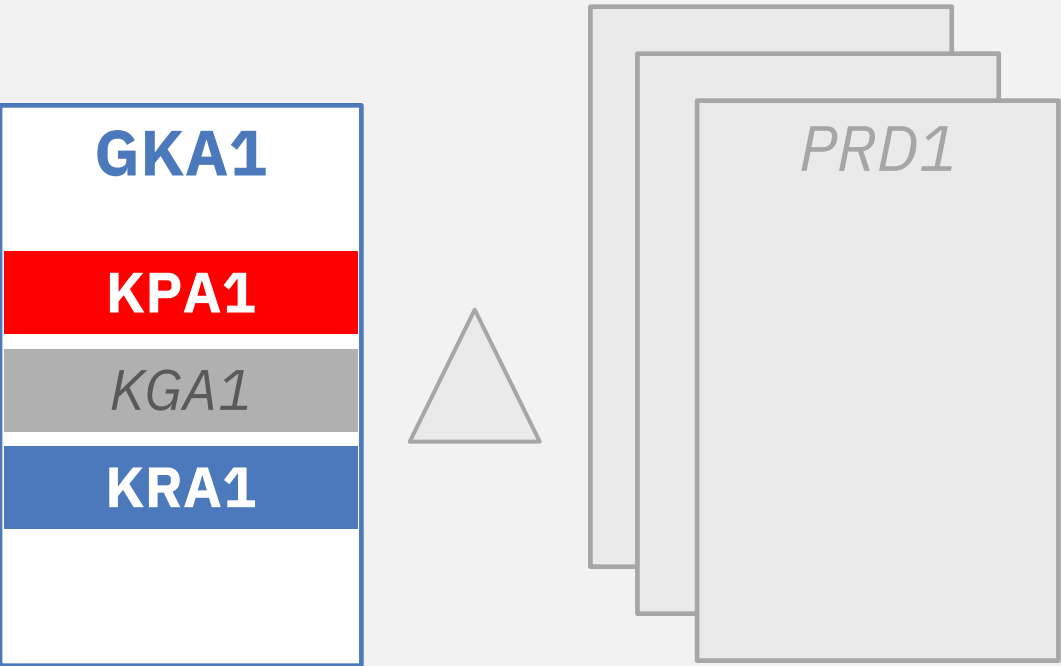
Metro Mirror



Planned region switch - MGM 4-site



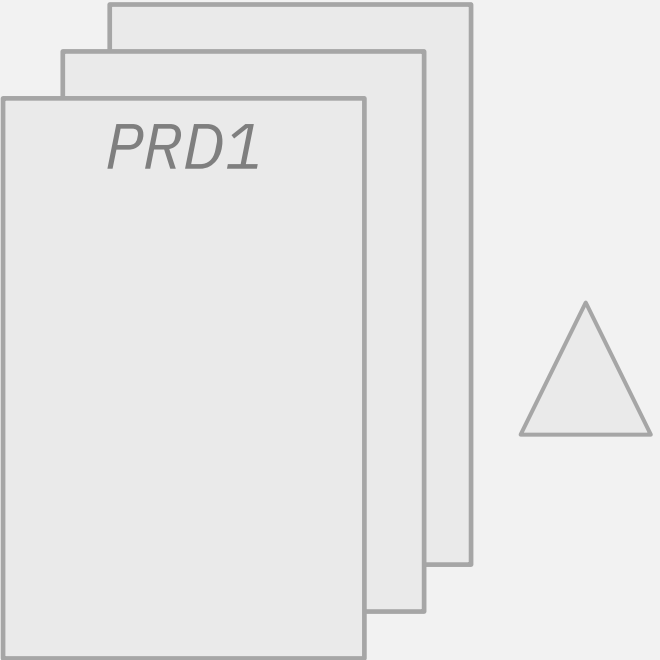
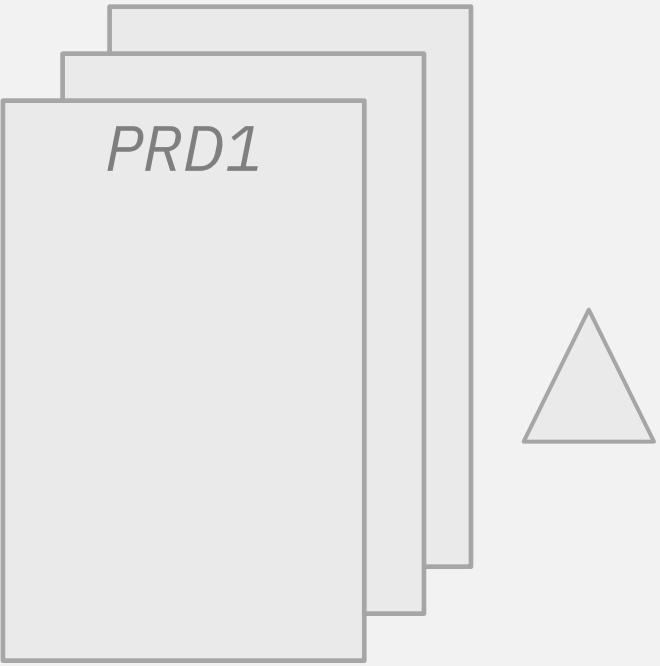
Region A



Global Copy



Region B



Metro Mirror



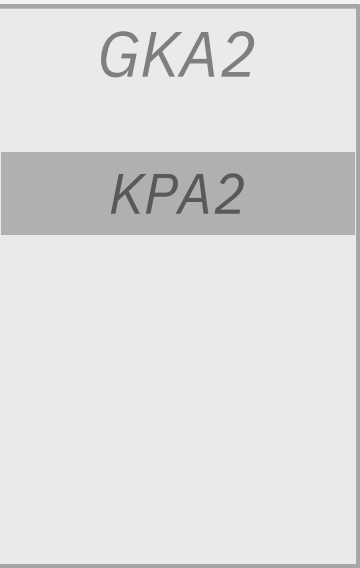
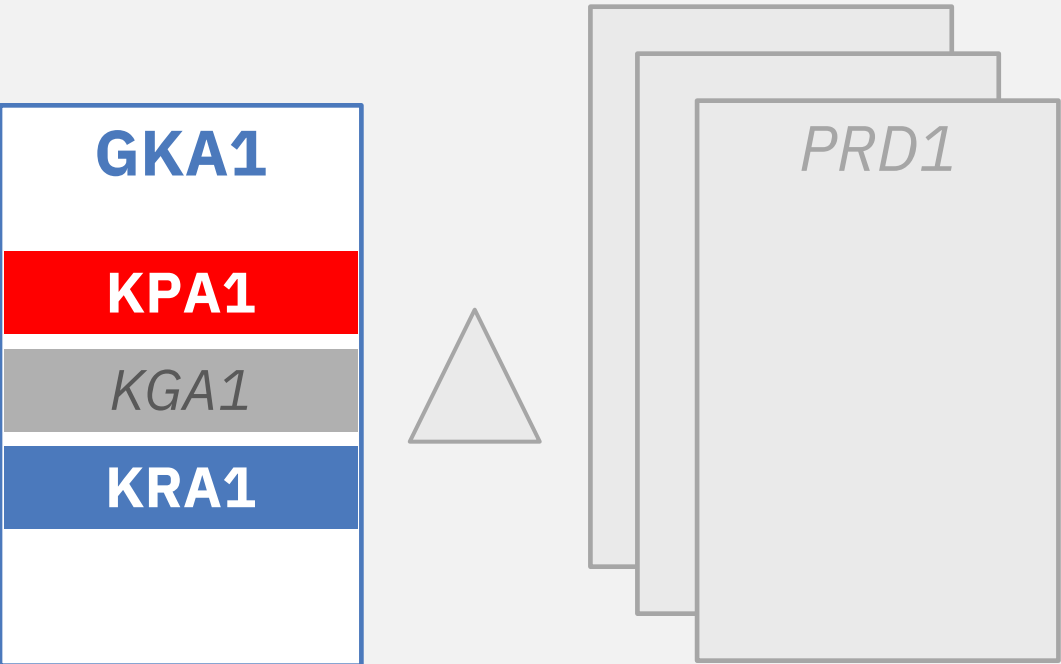
Bring down production, including KPA2
PROCEDURE= MGM4SITE_SWITCH_CAS_RS1
GDASD = 'INITIALIZE'
Bring up production, including KPB2

Global Mirror

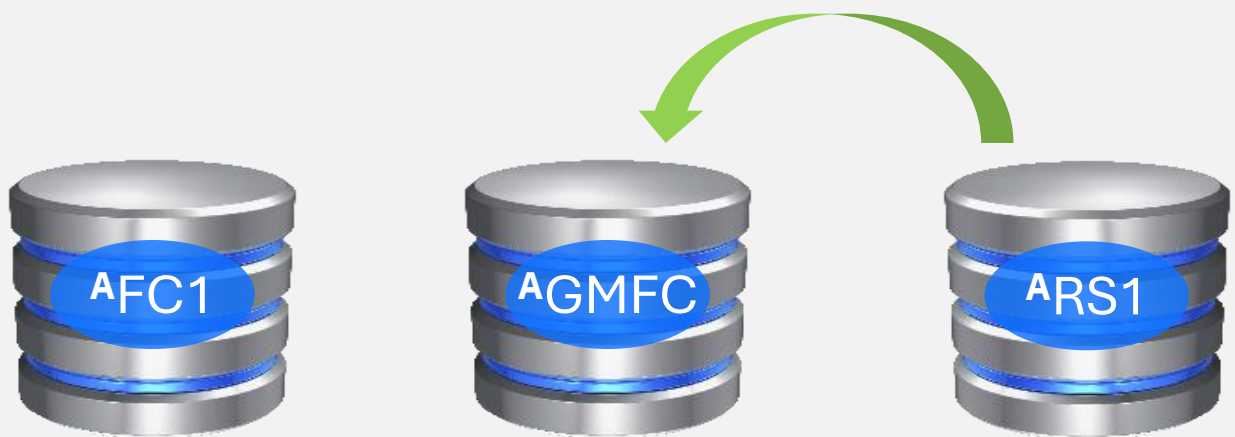
Planned region switch - MGM 4-site



Region A



Bring down production, including KPA2
PROCEDURE= MGM4SITE_SWITCH_CAS_RS1
GDASD = 'INITIALIZE'
Bring up production, including KPB2

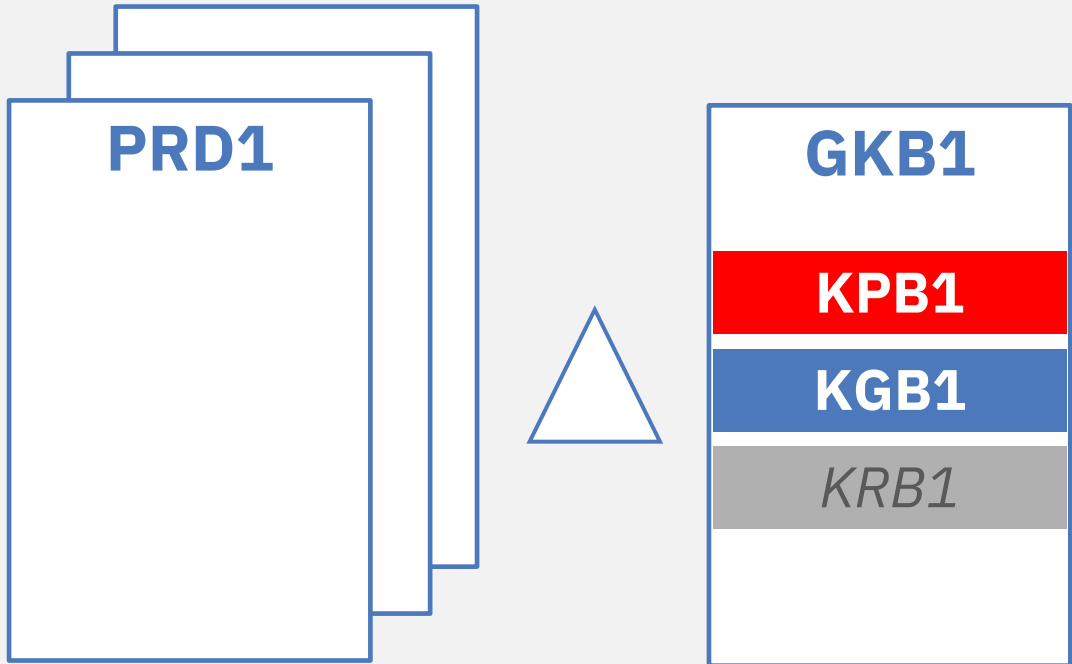


Global Copy



Global Mirror

Region B



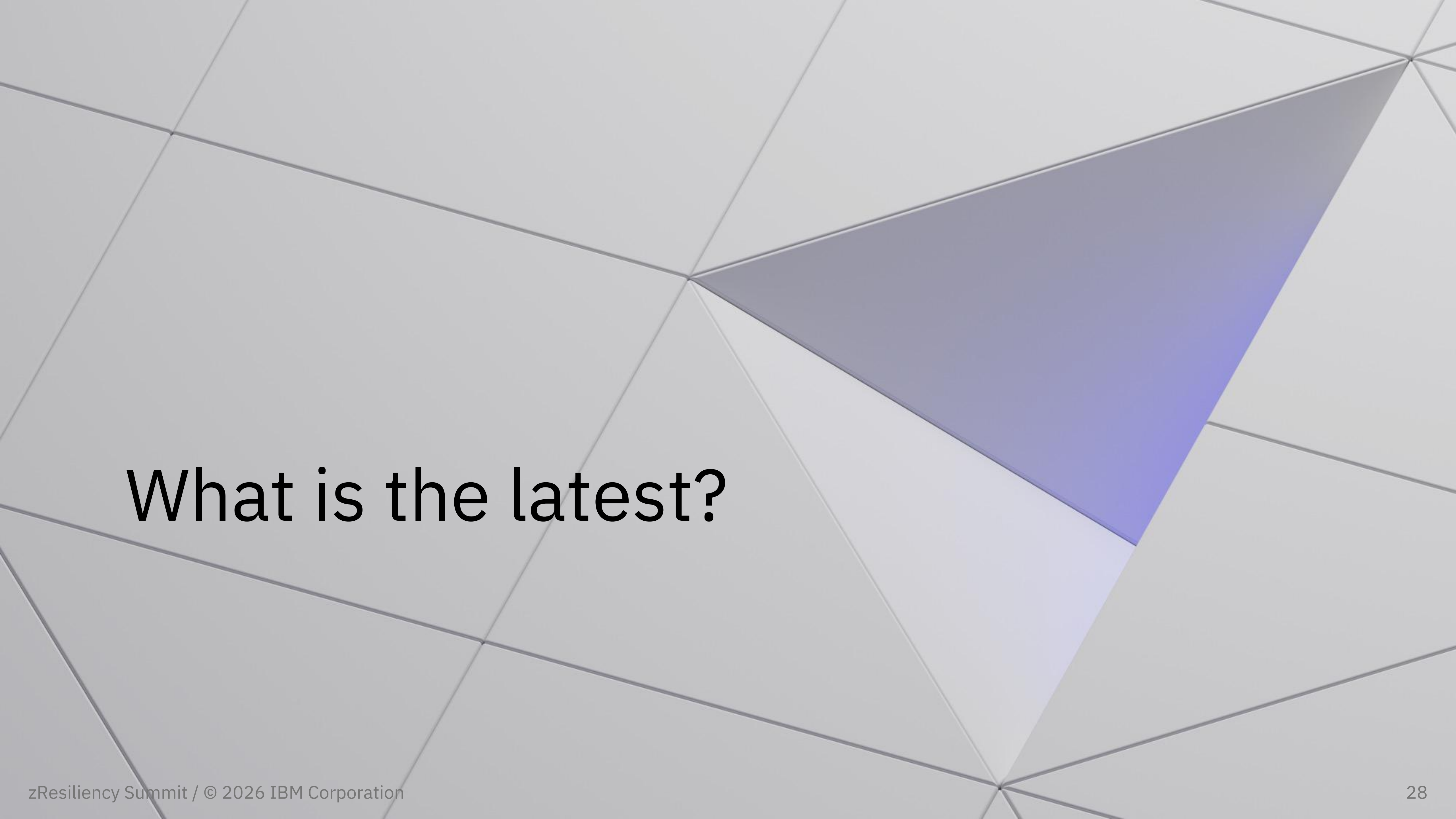
Metro Mirror



Powerful, including Procedure Handler, scripting capabilities

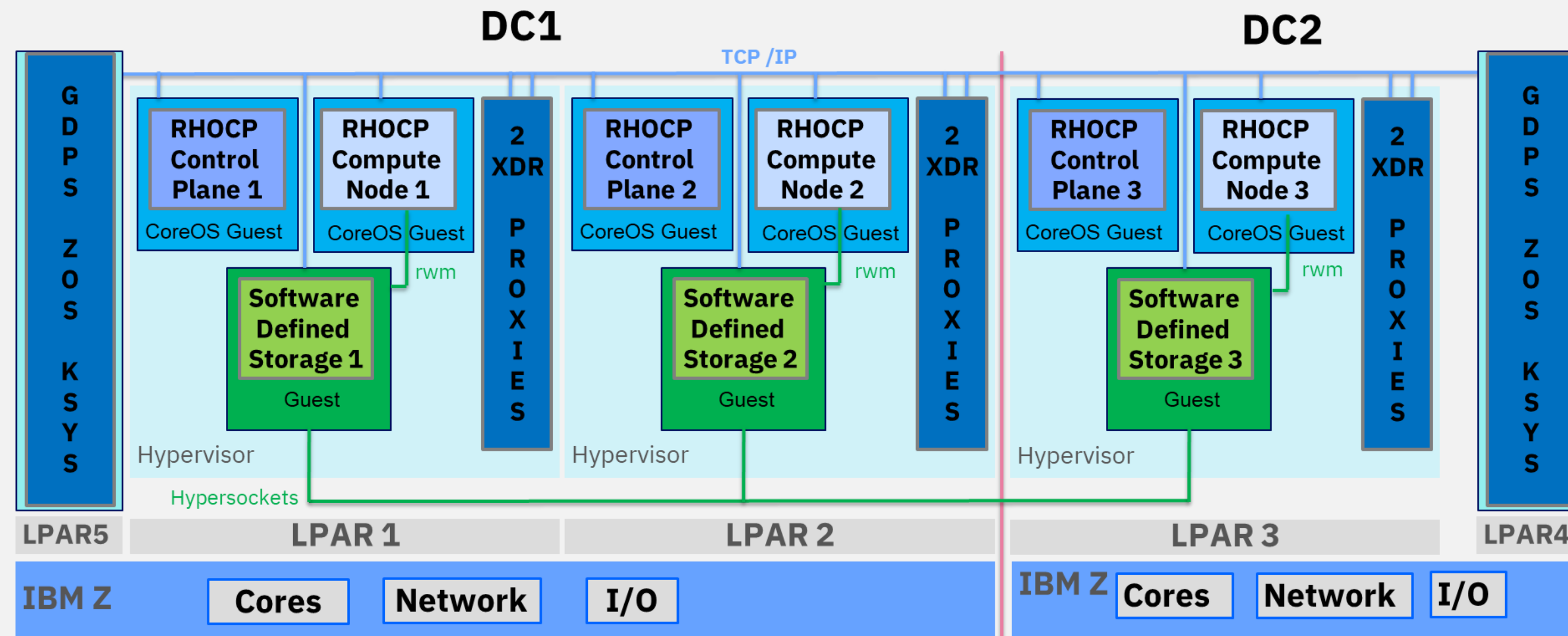


- Script statements and procedures to manage disk reconfiguration
 - Intelligence to query the environment to determine starting position (cascaded or multi-target)
 - Perform required actions based on that position
- Procedures are provided to support a number of scenarios, for example
 - Planned region switch and return home
 - Unplanned recovery in Recovery Region (real DR event) and return home
 - DR validation test on Recovery Region FC1 or X-disk
 - Move GM sessions incrementally when GM primary is lost

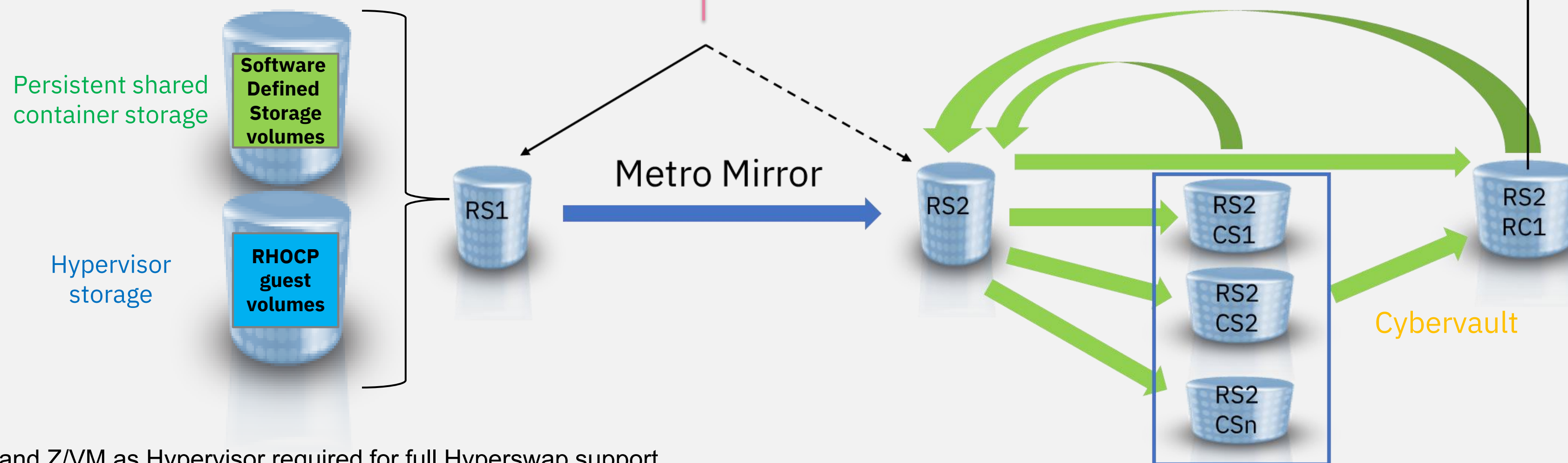
The background features a light gray field with a network of thin, dark gray lines intersecting at various points. A large, prominent triangle in shades of blue and purple is positioned on the right side of the frame, pointing towards the top right corner. The text "What is the latest?" is centered in the lower-left area of the image.

What is the latest?

GDPS can fully protect container infrastructure and workloads on IBM zSystems



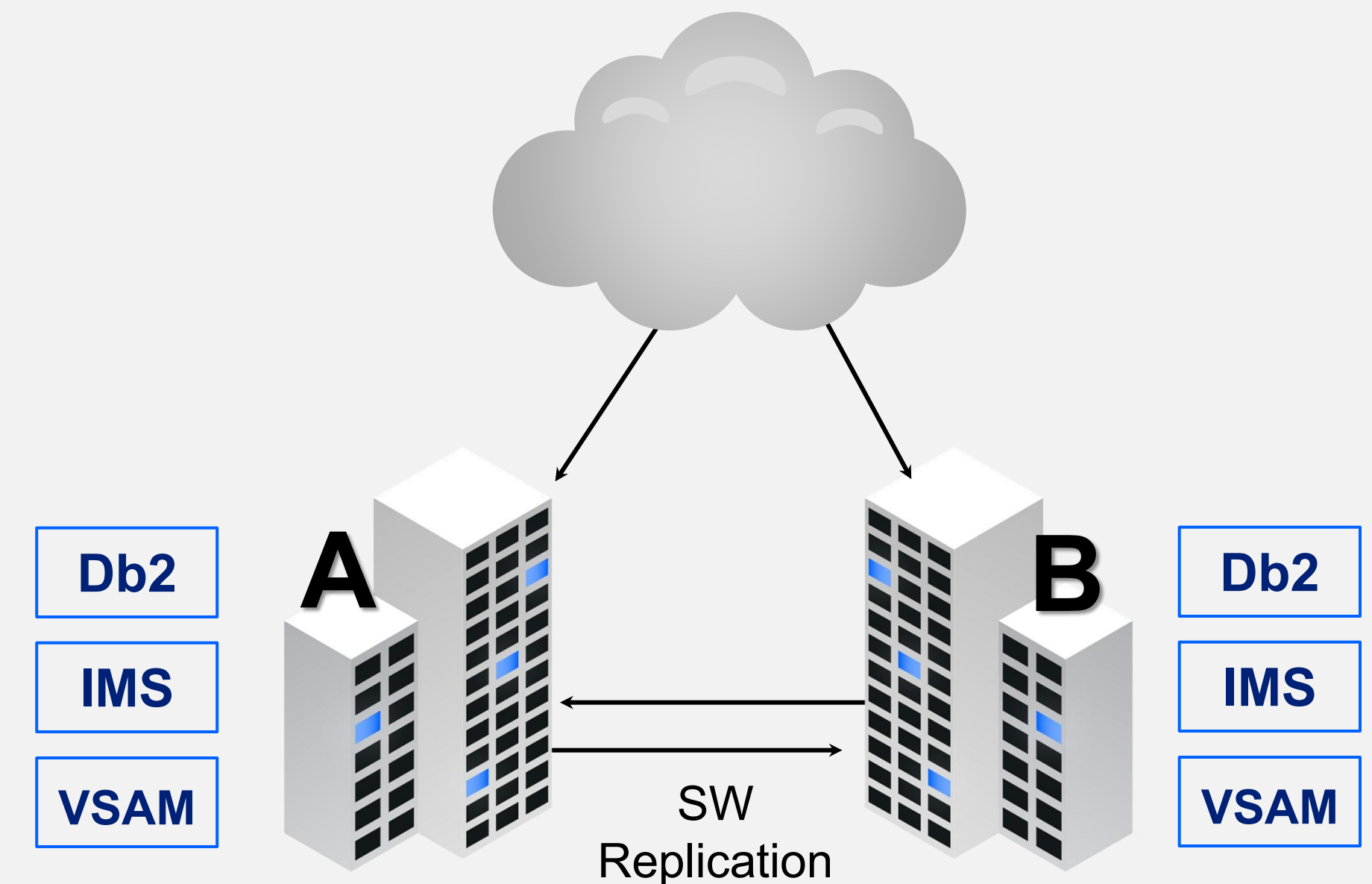
- GDPS can provide full HA / DR for containers and support storage replication of
- Hypervisor local storage (container infrastructure nodes)
 - Shared software defined storage (container required persistent storage)
 - Under Logical Corruption Protection (Cybervault)



- ECKD DASD volumes and Z/VM as Hypervisor required for full Hyperswap support
- z/VM SSI Live Guest Relocation driven by GDPS for optimized High Availability across several LPARS of Container infrastructure nodes, or Software Defined Storage nodes.
- GDPS tested with  and IBM Spectrum Scale as Software Defined Storage layer (IBM Spectrum Scale Container Native Storage Access (CNSA) on RHOC compute nodes)

GDPS Continuous Availability for OLTP workloads

- Manages availability at a workload level
- Provides a central point of monitoring and control
- Manages replication between sites
- Provides the ability to perform a controlled workload site switch
- Helps reduce recovery time and recovery point objectives – measured in seconds
- Provides near-continuous availability and helps simplify disaster recovery with an automated, centralized solution
- Facilitates better regulatory compliance management with a more effective business continuity plan
- Simplifies system resource management



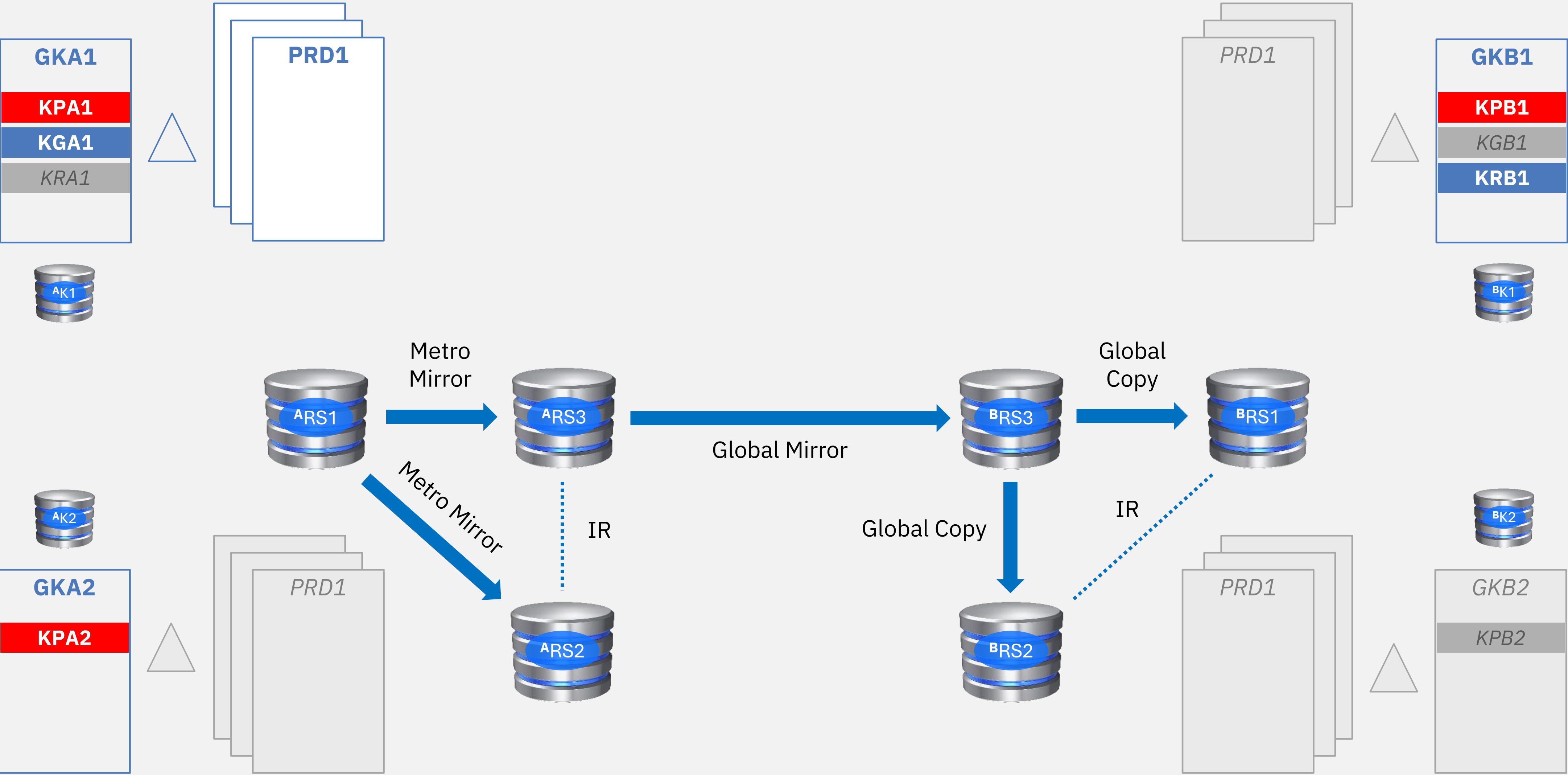
RPO = seconds RTO = 30 seconds or less

GDPS Metro Global – GM 6-site (aka MGM 6-site)



Region A

Region B

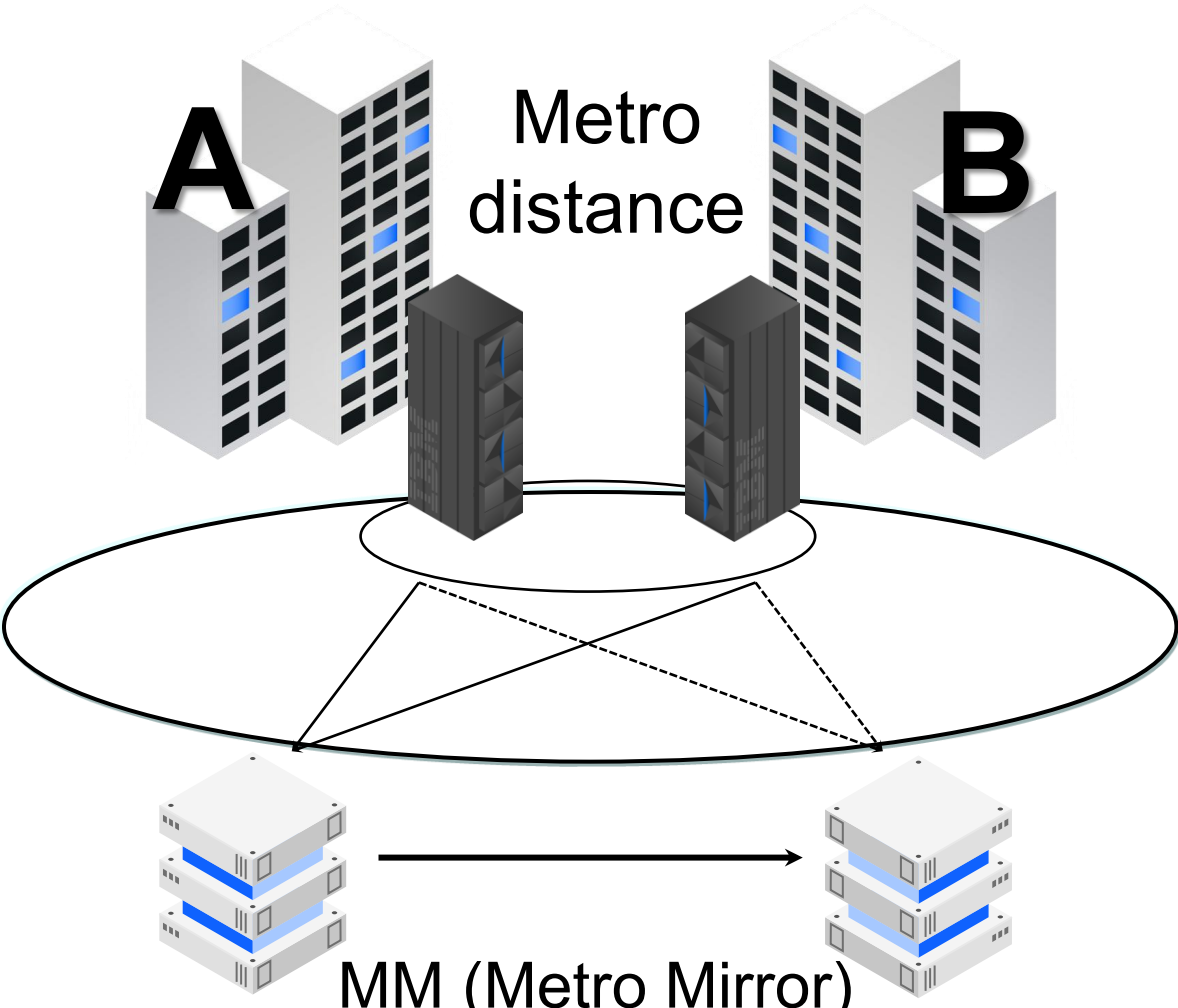
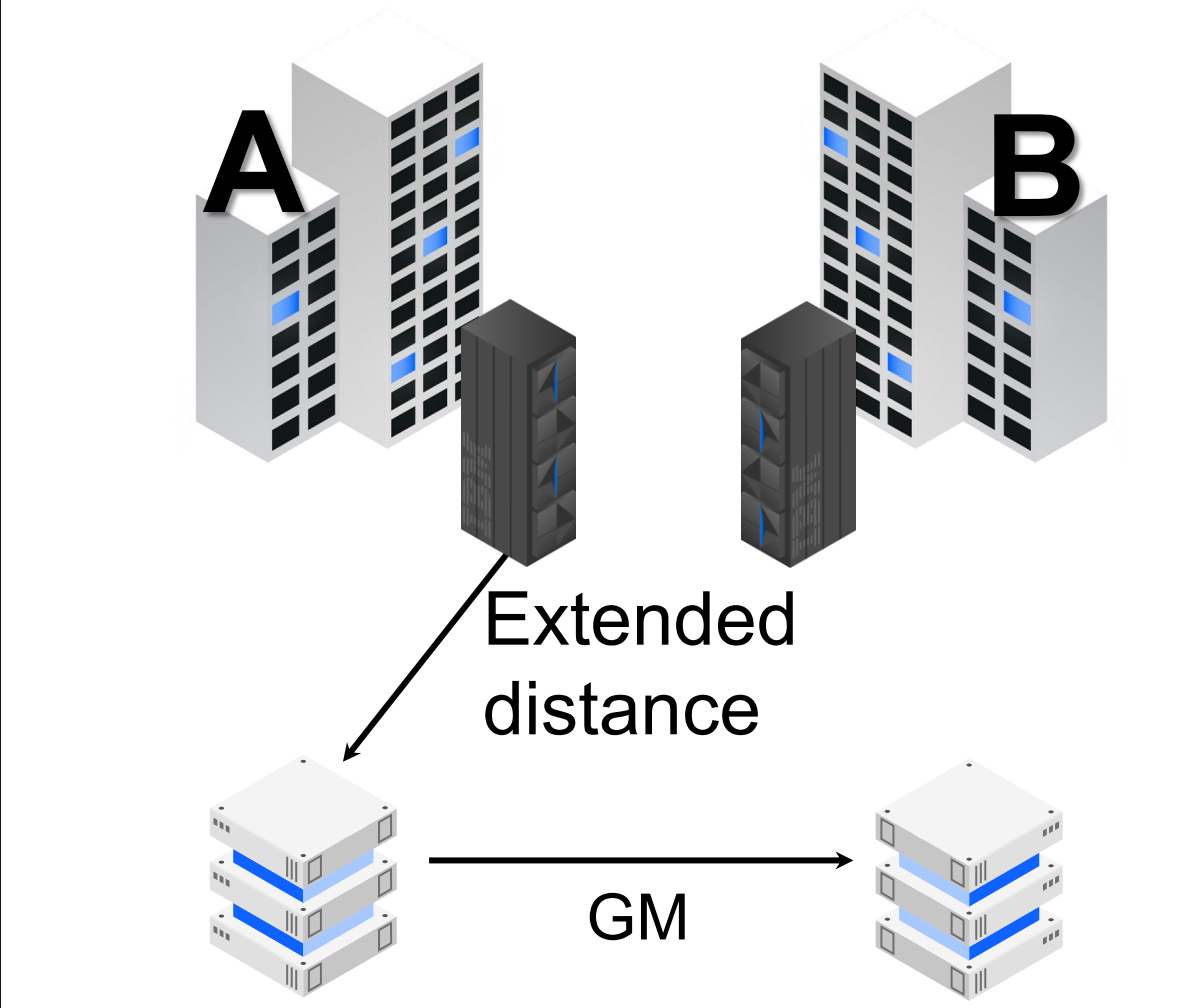
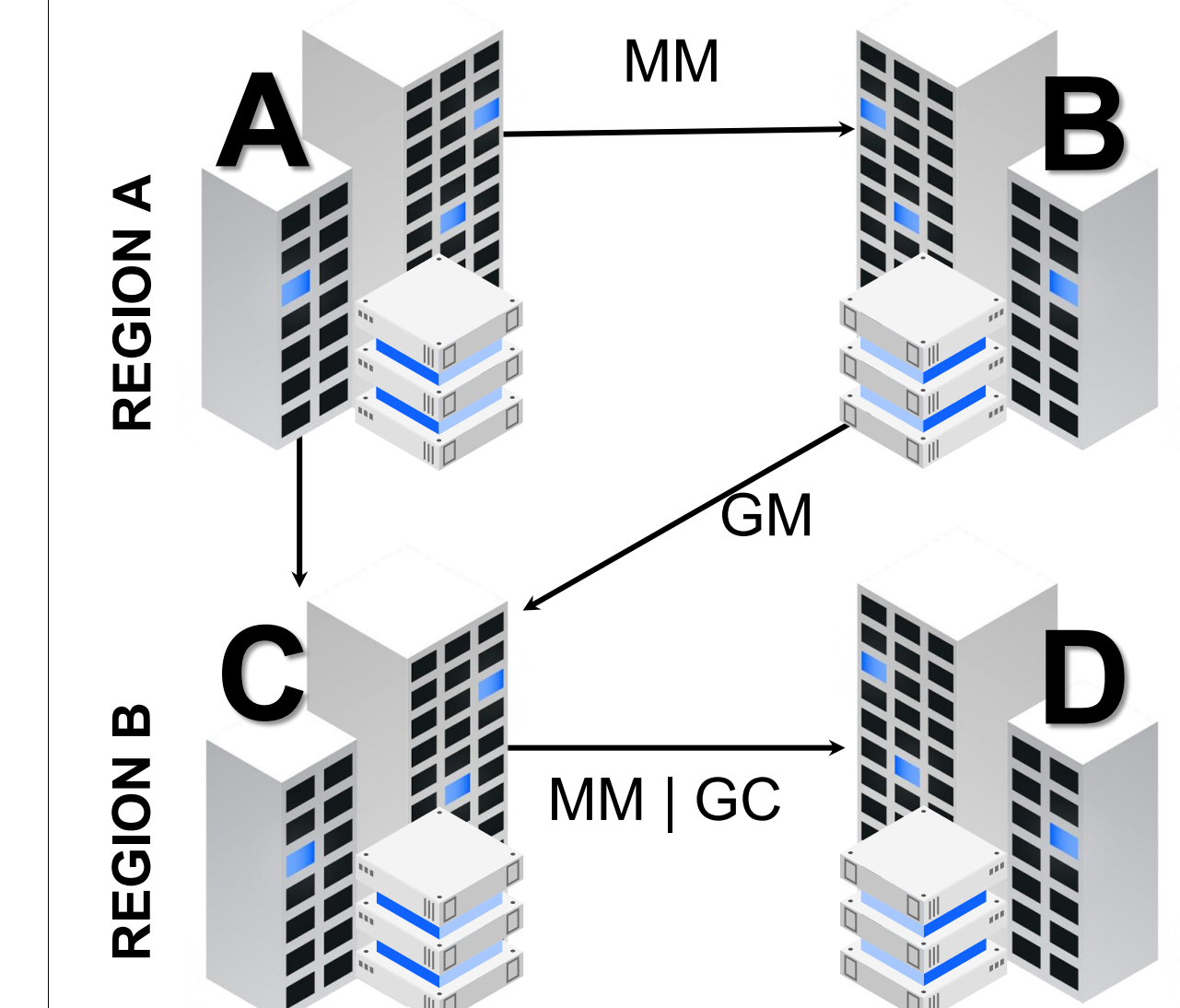
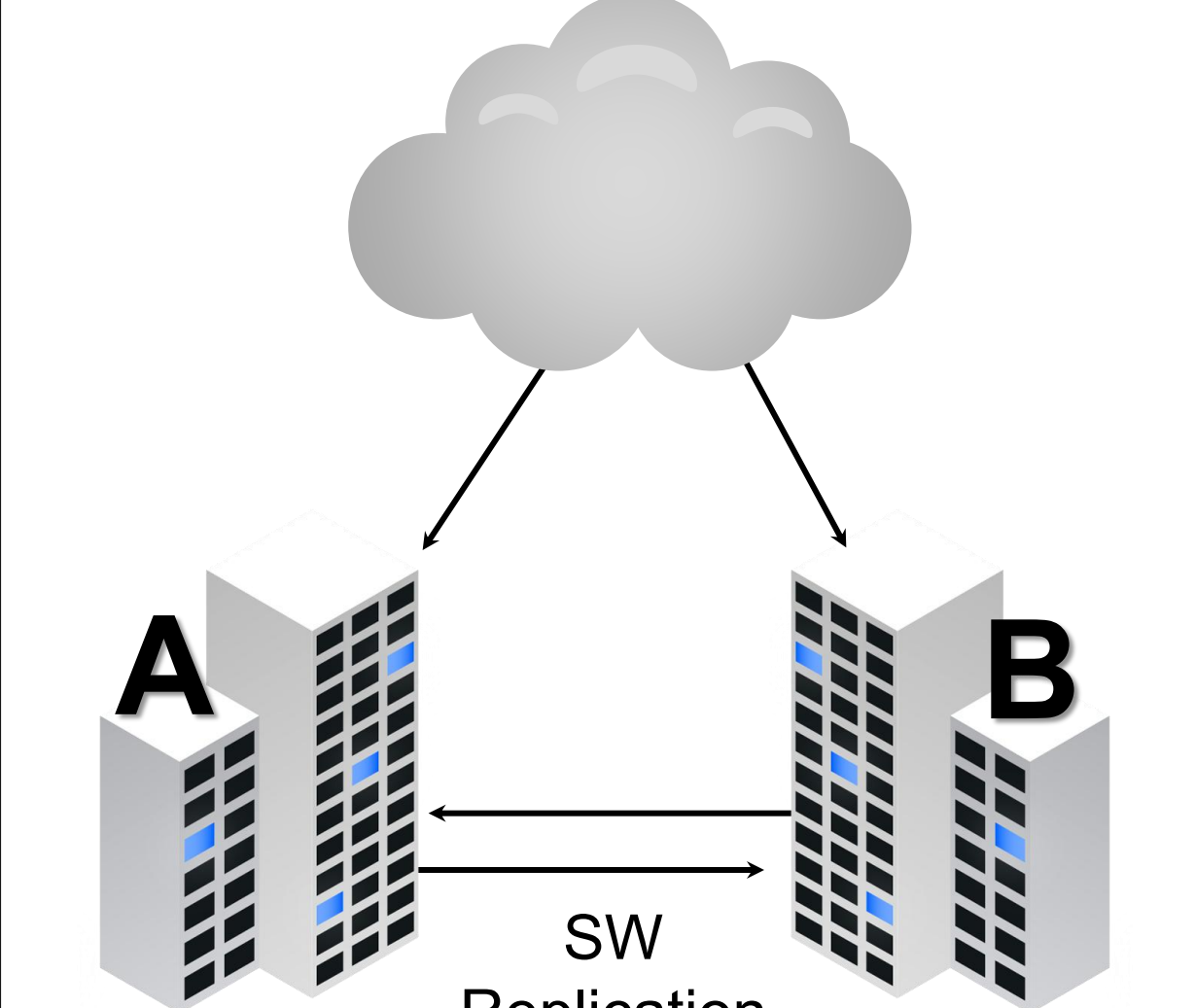




GDPS Products / Features and Capabilities

Balanced solutions (including features) designed to address different requirements



GDPS Metro	GDPS Global	GDPS Metro Global	GDPS Continuous Availability
Near-continuous availability and recovery at metro distances	Disaster recovery at extended distance	Near-continuous availability regionally & recovery for 3, 4 , & 6 sites	Near-continuous availability, recovery & workload balancing
Systems remain active Multisite workloads can withstand site and storage failures  RPO 0 & RTO <60 min	Rapid systems DR with “seconds” of data loss  RPO 3-5 sec & RTO <60 min	Metro near-continuous availability and out of region disaster recover  RPO 3-5 sec & RTO <60 min	Continuous availability at unlimited distances  RPO 3-5 sec & RTO <60 sec

Balanced solutions (including features) designed to address different requirements



- **z/OS proxy** – provides CA / DR for mono-plexes & sysplexes and disk sharing with systems outside the sysplex
- **Logical Corruption Protection (LCP) Mgr** – ability to take several FC / SGC point-in-time (PiT) copies of data & manage them, verify PiT copies, and, in the event of a cyber event, restore PiT copies to production this is the underlying infrastructure for the IBM Z Cyber Vault and used by several GDPS clients
- **Test Copy Manager** – creates a consistent copy of data for testing (can be used in conjunction with zBUrST for application stress testing)
- **SSC / IDAA** - provides CA / DR for IDAA instances
- **Dual Leg** – provides two synch legs for maximum metro CA / DR
- **GDPS GM2SITE Bi-directional** - provides region switch enhancements including the capability to independently switch the replication direction of GM sessions where multiple sessions (or consistency groups) are being managed in a single GDPS instance.
- **GDPS Continuous Availability Zero Data Loss (ZDL)** – provides no data loss capability at metro and unlimited distance topologies
- **zKVM** – provides CA / DR for zKVM instances
- **GDPS Solution Manager (GSM)** – reduces the number of k-sys for clients with multiple GDPS MGM4SITE or GDPS MGM6SITE configurations

Each GDPS solution provides extensive capabilities.



- **User interface** – 3270 & GUI and enterprise portal ...more mature people tend to use the 3270 UI and younger the GUI
- **OS / Lpar / CECs mgt** – ACTIVATE / DEACTIVATE / LOAD Ipars (including CF & SSC), START / STOP OS instances, temporary capacity ACTIVATE / UNDO, and CTN configuration ... large majority of GDPS clients use GDPS “Standard Action” to operate their mainframe environment
- **Replication mgt** – start, stop, resynch, & reconfiguration for site / region switching / synch & asynch disk replication ... all GDPS clients use GDPS “DASD Management” to manage DS8K MM synch replication, GM asynch replication, & FC
- **Sysplex mgt** – CDS & CF (including structures) ... several GDPS clients use GDPS “Sysplex Resource Management” to manage their PS
- **Workflows** – orchestrates a set of actions to provide repeatable results (e.g., planned / unplanned site / region switch) ... large majority of GDPS clients use GDPS “Planned Actions” to fully automate planned / unplanned site / region switching

Each GDPS solution provides extensive capabilities (cont).



- **Monitoring & Health Checks** – verify that the environment is healthy and optimized for best practices ... all GDPS clients use monitoring and the large majority execute GDPS Health Checks
- **RESTful API** – programatically pass policy to GDPS, tell GDPS what to do, and get status from GDPS ... clients starting to exploit the RESTful API

Summary

What value can the IBM GDPS solution offer your organization?

Experience	Commitment		Value		Vision
Client acceptance –Almost 1400 GDPS licenses installed in 51 countries worldwide –Tested technology to support automated and repeatable results –Complete implementation guided by experienced consultants	Open industry standards –IBM GDPS supports industry-accepted, open replication architectures (Metro Mirror, Global Mirror and Fibre Channel) –Architectures licensed by all enterprise storage vendors –GDPS qualification program for GDPS Metro (IBM, Hitachi)	Investment protection –Designed to be easily upgradeable –Common code base for each product	Product maturity –Generally available since 1998 –Suite of products –Enterprise-to-enterprise capability –Many years of IBM Z production experience –CA and DR best of breed –Continually enhanced	Client focus –GDPS Design Council –Synergy with IBM development labs –Incorporates several IBM patents –New release planned every year –GDPS advocate program –GDPS Requirements process	IBM support –Fully supported via standard IBM support structure –Fixes through normal IBM Z channels

Backup and resources

Additional Information

■ Web sites:

- GDPS <https://www.ibm.com/products/gdps>
- IBM zSystems <https://www.ibm.com/z>
- IBM zSystems Resiliency <https://www.ibm.com/z/resiliency>
- Storage <https://www.ibm.com/storage>
- Redbook – GDPS Family: An Introduction to Concepts and Capabilities
<http://www.redbooks.ibm.com/abstracts/sg246374.html?Open>

■ GDPS Web site resources

- GDPS: The Enterprise Continuous Availability / Disaster Recovery Solution white paper
- GDPS pre-requisite information
- GDPS training schedule links
- GDPS hardware qualification letters
- E-mail: gdps@us.ibm.com



ITSO Redbooks

[IBM GDPS: An Introduction to Concepts and Capabilities](#)

This IBM Redbooks® publication presents an overview of the IBM GDPS® offerings and the roles they play in delivering a business IT resilience solution. The book begins with general concepts of business IT resilience and disaster recovery (DR), along ...

[Getting Started with IBM Z Cyber Vault](#)

Cyberattacks and data corruption threats continue to escalate, which makes cyber resiliency a critical component of an enterprise IT strategy. Although traditional disaster recovery (DR) focuses on physical failures, cyber resiliency addresses logical ...

[Getting Started with Business Continuity on IBM Z](#)

IBM Z® platforms underpin mission-critical enterprise workloads where disruptions directly impact availability, data integrity, and business outcomes. Business continuity is achieved through engineered resiliency and recovery, leveraging platform ...

[Getting Started with z/OS Data Set Encryption](#)

This IBM® Redbooks® publication provides a broad explanation of data protection through encryption and IBM Z® pervasive encryption with a focus on IBM z/OS® data set encryption. It describes how the various hardware and software components interact in a ...

[Transitioning to Quantum-Safe Cryptography on IBM Z](#)

As cyberattacks continue to increase, the cost and reputation impacts of data breaches remain a top concern across all enterprises. Even if sensitive data is encrypted and is of no use now, cybercriminals are harvesting that data because they might gain access to a quantum computer that can break classical cryptographic algorithms sometime in the future ...